Manipulating Temporal Expectations to Accelerate Recovery After Cataract Surgery: A Randomized Controlled Trial

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Abstract

Perceived time can objectively influence physiological outcomes, but such effects have been produced using false temporal feedback in laboratory settings, which is difficult to extend clinically. Drawing on the Embodied Models of Health (EMH) framework, in which beliefs shape health by orienting attention and expectation, the present study tested whether shifting patients' temporal expectations could influence post-surgical recovery. Patients undergoing bilateral cataract surgery (N = 147) were randomly assigned to a fast healing condition, comprising a preoperative video emphasizing faster-than-standard recovery timelines and daily surveys directing attention toward healing variability, or a standard healing condition, comprising conventional information and surveys unrelated to healing. Visual acuity was assessed at baseline and at 1-day, 1-to-2-week, and 3-month follow-ups. Group-level differences across conditions were directionally consistent but modest. The theoretically motivated analyses revealed an Attention × Sensitivity interaction consistent with the framework's core prediction: among patients who absorbed the faster expected healing times, greater daily attention to healing variability predicted faster recovery; among those who did not, attention had little effect. By the morning after surgery, and at every follow-up through three months, high-attention patients with strong healing expectations or high sensitivity to the faster-healing message held an acuity advantage over belief-matched controls equivalent to reading print roughly 10% smaller at the same distance, and reported at least 50% greater improvement in self-reported visual function. These findings provide the first clinical evidence that attention functions as a belief-weighted mediator between psychological and physiological processes, extending laboratory demonstrations of mind-body unity to clinical practice.

Keywords: expectations, placebo effects, mind-body unity, attention, cataract surgery, visual acuity, randomized controlled trial

Public Significance Statement

This randomized controlled trial shows that telling cataract surgery patients to expect faster healing, and asking them to attend daily to small changes in their vision, was associated with objectively faster recovery of visual acuity among patients who absorbed the faster timelines. The findings suggest that how clinicians frame recovery timelines is not merely informational but may itself influence the pace of healing.

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In 2023, Aungle and Langer published a study that found that experimentally induced wounds objectively healed faster the more time participants believed had passed, even though the true elapsed time was always the same (Aungle & Langer, 2023). Their paper is part of a relatively nascent field of inquiry within mind-body research that demonstrates perceived time has real physiological consequences independent of real time. Researchers have also found that perceived time independently influences blood glucose levels (Park et al., 2016), perceived sleep independently influences brain activity and cognitive function (Rahman et al., 2020), and perceived time independently predicts neuromuscular fatigue (Matta et al., 2025; Matta et al., 2024). In all such studies, researchers use false temporal feedback to produce the reported effect – an approach that is hard, if not impossible, to extend beyond the laboratory.

One way to think about the underlying mechanisms across these studies, however, is to think of perceived time as an input to expectation formation, with expectations ultimately producing the effects. We know from the voluminous literature on placebo effects that expectations are a fundamental driver of physiological functioning (Finniss et al., 2010; Geuter et al., 2017; Price et al., 2008; Wager & Atlas, 2015), so it stands to reason that effects similar to those produced by perceived time can be produced by shifting temporal expectations. *In other words*, *if we change how quickly someone expects a physiological process to unfold, we should be able to influence how quickly it actually happens.* That is the premise underlying the current study, which aimed to test whether manipulating patients' healing time expectations following cataract surgery actually changes how quickly they heal. This question builds on a large body of work on placebo effects.

Placebo effects, expectations, and health

For centuries, we have known that expectations and health are inextricably linked (De Craen et al., 1999; Finniss et al., 2010; Kaptchuk, 1998). Rigorous scientific study of this process, however, only got fully underway after Henry Beecher "discovered" the placebo effect and introduced randomized controlled trials (RCTs) to medical science (Beecher, 1955). Working as a medic during World War II, Beecher ran out of morphine while caring for injured soldiers, so he decided to pretend he was injecting morphine when in reality he was using saline. Much to his surprise, many of the soldiers responded to the saline as if it was morphine: they experienced real pain relief because they expected an analgesic effect. Beecher was not the first to observe that expectations fundamentally shape health – for example, Wolf (1950) famously told a 28 year old pregnant woman suffering from nausea that ipecac, a medication used to *induce* vomiting, would *alleviate* her nausea, which it then did – but he was the first to systematically document the phenomenon, quantify its effects across clinical contexts, and argue persuasively that patients' expectations should be treated as a central variable in medical science.

Beecher's work led directly to the study of "the pharmacology of placebo," which established that placebo treatments share many of the characteristics of actual drugs – e.g., peak effects, cumulative effects, carryover effects, and efficacy inversely related to severity (Lasagna et al., 1958). For the next two decades, although substantial research on placebo effects was conducted, the idea that they were a kind of subjective illusion remained (Enck et al., 2013; Kaptchuk & Miller, 2015). That started to change in meaningful ways as a result of three new developments in the field. One, researchers began to understand the neurobiology of placebo effects (Levine et al., 1978), which demonstrated that placebos work through biological mechanisms (i.e., that they are not subjective illusions). Two, Ader and Cohen demonstrated that conditioned expectations alone can shape immune function, giving rise to the field of psychoneuroimmunology (Ader & Cohen, 1975). Three, Langer and colleagues published several studies that underscored how simple differences in the ways we think can lead to substantial improvements in longevity and health (i.e., that the “placebo” phenomenon extends well beyond doctors’ offices and hospitals) (Langer et al., 1990; Langer et al., 1975; Langer & Rodin, 1976; Rodin & Langer, 1977).

These three developments led to a proliferation of new research that made clear expectations about our health significantly shape the health we experience. Irving Kirsch’s response-expectancy theory helped shift thinking to expectations as central mechanisms in placebo responses (Kirsch, 1985). Benedetti and colleagues uncovered diverse mechanisms underlying the effects of expectations on physiology (Benedetti & Amanzio, 1997) and demonstrated that hidden administration of active treatments is less effective (Benedetti et al., 2003), underscoring the central role played by expectations beyond inert treatments. While much of the early work involved mental health and pain, de la Fuente-Fernandez and colleagues showed that simply expecting medication for Parkinson’s disease causes the brain to release a significant amount of dopamine, even though the patients only received a placebo injection (de la Fuente-Fernández et al., 2001). Moseley and colleagues found that sham knee surgery produced the same benefits as real knee surgery, demonstrating that expectations are fundamental to outcomes from invasive procedures as well (Moseley et al., 2002). Crum and Langer showed that simply expecting the benefits of physical exercise can produce those benefits, even when behavior remains unchanged (Crum & Langer, 2007). Kaptchuk and colleagues established that “open-label placebos” – placebos given to patients who know they are taking a placebo – can still produce real health benefits (Kaptchuk et al., 2010), providing evidence that placebo effects do not depend entirely on deception.

As the underlying science has advanced, and the breadth of psychological influences on physical health has become clearer, the stakes surrounding the principle of mind-body unity, first articulated in Langer et al. (1990), have grown. In previous work, we argued that the term “mind-body connection” is far more common than “mind-body unity,” and that this subtle semantic difference obscures the breadth of psychological influences on health (Aungle & Langer, 2023). We recently proposed an Embodied Models of Health (EMH) theoretical framework for understanding how mind-body unity works in practice (Aungle, Matta, Chen, et al., 2026), which we formalized into a computational model in a complementary companion paper (Aungle, Matta, Loecher, et al., 2026); within the EMH framework, beliefs about how health works can become self-fulfilling by shaping attention allocation and expectation formation processes. Expectations are fundamental drivers of mind-body effects, and one way to understand the perceived time findings mentioned at the beginning of this paper is to think of perceived time as an input to expectation formation.

The Present Study

Given that the perceived time studies mentioned earlier occurred under the tightly controlled conditions of laboratories, a natural question arises – how can we extend these findings to real life, outside the lab? As we argued above, one way to do so could be to intervene on temporal expectations directly. In the present study, we explored this possibility in the context of patients recovering from cataract surgery.

Cataract surgery is one of the most common and standardized surgical procedures worldwide, with a highly predictable recovery trajectory and well-characterized time course of visual improvement (Grzybowski & Kanclerz, 2019). Patients are routinely given explicit information about what to expect during recovery, making healing timelines both salient and clinically relevant (e.g., National Health Service, 2025). At the same time, recovery unfolds over days to weeks – long enough for expectations to plausibly shape physiological processes, but short enough to measure objective outcomes with precision. The popularity of the procedure, its standardized nature, and the predictable recovery trajectory were all reasons we chose to focus on recovery from cataract surgery.

Past research has found that expectations independently and significantly shape surgical outcomes. For example, in a study of patients undergoing coronary artery bypass graft (CABG) surgery, researchers randomly assigned participants to either a pre-surgery intervention designed to optimize outcome expectations, an active control intervention focusing on emotional support and advice, or a standard medical care (SMC) condition that proceeded as usual. The patients in the outcome expectations condition showed significantly larger improvements in disability compared to the SMC condition and marginally significant improvements compared to the active control condition (Rief et al., 2017). There is also evidence that temporal expectations can independently shape physiological outcomes. For example, in a study of placebo analgesia, researchers told some participants that the (placebo) analgesic cream takes 5 minutes to start working, some participants that it takes 30 minutes to start working, and some participants that it was simply a cream with hydrating properties; the analgesic effect of the placebo tracked the expected time of onset (Camerone et al., 2021).

We therefore conducted a randomized controlled trial to test whether experimentally manipulating healing time expectations following bilateral cataract surgery influences actual healing times. We designed the intervention with our Embodied Models of Health framework in mind (Aungle, Matta, Chen, et al., 2026), in which beliefs are the fundamental drivers of mind-body effects on health as a result of their influence on attention allocation and expectation formation. Because beliefs and expectations often go hand in hand, the intervention was two-pronged. The first prong, a preoperative information video, targeted the beliefs and expectations components of the framework by shifting the healing timeframes participants were given. The second prong, a series of daily check-in surveys, targeted attention allocation by prompting participants to notice subtle day-to-day changes in their recovery. In the EMH framework, attention is a belief-weighted mediator between mind and body: the video was intended to shape how quickly participants believed and expected healing to unfold, and the daily surveys were intended to amplify healing physiology consistent with those expectations. Assuming participants in the fast healing condition internalized the expected healing times they were given, noticing variability in healing consistent with those expectations should facilitate actual healing.

The intervention was, by design, quite minimal. Participants in the fast healing condition could easily come across the standard healing timeframes for cataract surgery online, undermining the information provided in the video, and could complete the check-in surveys perfunctorily, reducing the influence of attention on the healing process. We also used an active control condition, matched in contact time and survey burden, to provide a conservative test. Any improvement in the fast healing condition relative to the standard healing condition would thus be strong evidence that psychology shapes the physiology of healing.

Hypotheses

Our primary hypothesis is that participants in the fast healing condition will show greater improvement in visual acuity (i.e., larger decreases in logMAR from baseline) at the 1-day and 1-to-2-week postoperative visual acuity exams compared to participants in the standard healing condition. Because visual acuity is measured at multiple post-operative time points — 1 day, 1–2 weeks, and 3 months after surgery — we use seemingly unrelated regression (SUR) models to jointly test this hypothesis across waves, allowing us to evaluate whether the intervention effect is jointly significant across the recovery period while accounting for the correlated structure of repeated observations on the same patients. Given our focus on healing, we center our models on visual acuity at the 1-day and 1–2-week time points. We use the 3-month data to assess effects on overall outcomes, once fully healed.

We have two further hypotheses motivated by our Embodied Models of Health (EMH) framework:

1. **Attention-to-healing hypothesis.** Among participants in the fast healing condition, daily attention to variability in the healing process (as measured by the average daily check-in survey scores) will predict larger improvements in visual acuity, even after controlling for age and baseline psychosocial variables (perceived stress, mindfulness, and anxiety). We test this using SUR to jointly evaluate the attention coefficient across post-operative waves.
2. **Sensitivity moderation hypothesis.** The effect of attention to healing on visual acuity recovery will be moderated by participants' sensitivity to the healing-time information provided in the pre-operative video, operationalized as the mean number of healing-time questions answered correctly. We predict a significant Attention × Sensitivity interaction, such that the relationship between attention and healing will be strongest among participants who internalized the expected healing times conveyed in the video.

Method

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. The study was approved by the Harvard University Committee on the Use of Human Subjects (IRB23-1420) and was not preregistered. Analyses were conducted in R and Stata.

Participants

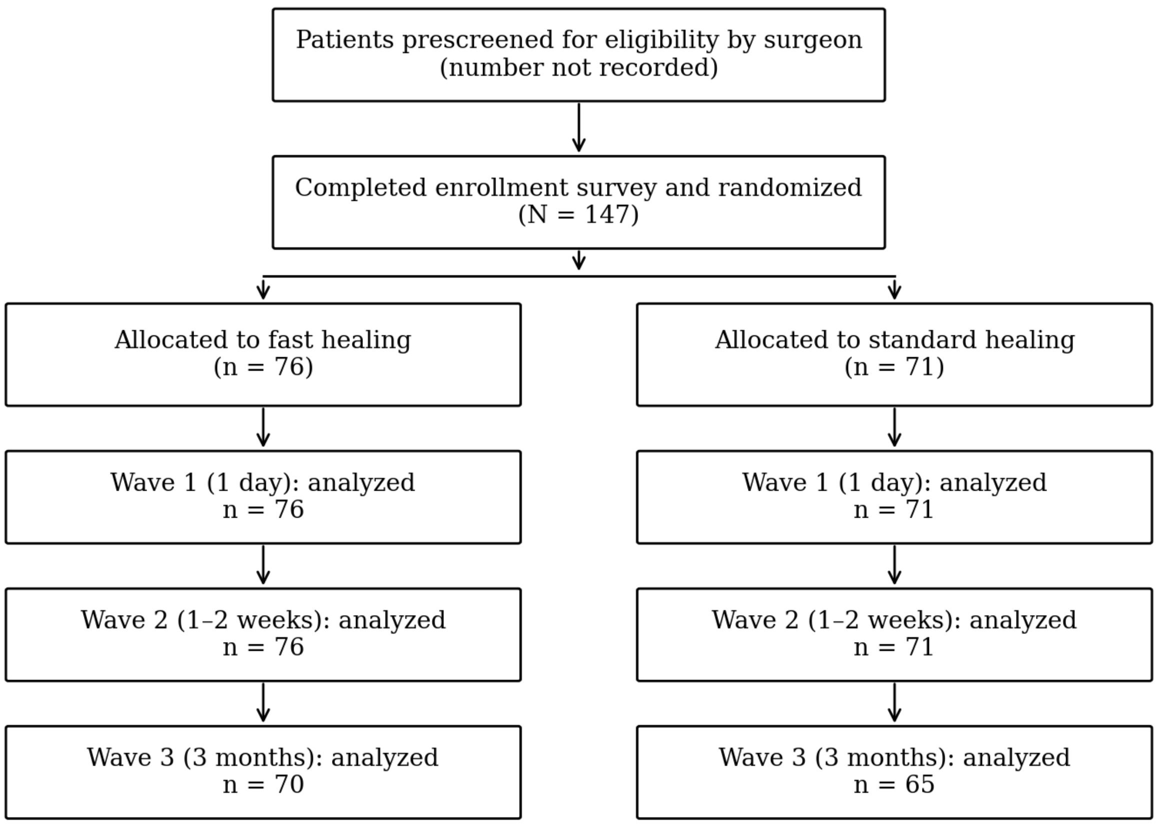
Participants were 147 adult patients scheduled for bilateral cataract surgery, recruited between late 2024 and early 2026 from two ophthalmology clinics in Vermont and Massachusetts for a study on "psychological influences on recovery following cataract surgery." All participants were operated on by the same surgeon. Participants averaged 71.5 years of age (*SD* = 7.0; range 43–84); 90 (61.2%) were women; and 94.6% identified as White, 3.4% as Asian, 0.7% as Black or African American, and 1.4% as another race or ethnicity.

Eligibility was determined by the surgeon prior to referral. Patients were eligible if they were scheduled for bilateral cataract surgery and were not exceptionally severe or complicated cases; patients with significant pre-existing, vision-limiting retinal or corneal pathologies were excluded at this prescreening stage. Because prescreening was conducted by the clinical team before patients were referred to the study, the number of patients considered but not referred was not recorded. All referred patients who completed the enrollment survey were enrolled and randomized, and no participants were excluded from the Wave 1 and Wave 2 analyses. One participant in the fast healing condition suffered a retinal detachment after surgery, a serious complication unrelated to the study; his 3-month visual acuity is excluded from all Wave 3 analyses. Figure 1 presents participant flow through the study.

An a priori power analysis conducted during study planning targeted a sample of approximately 180 participants, which provides 80% power to detect a between-group difference of $d \approx 0.37$ with a one-tailed test at $\alpha = .05$, consistent with our directional hypotheses ($d \approx 0.42$ two-tailed). Because recruitment proceeded more slowly than anticipated, enrollment was closed after 147 participants; this decision was based on the pace of recruitment and was made before any outcome data had been examined. A sensitivity analysis indicates that the achieved sample provides 80% power to detect $d = 0.41$ with a one-tailed test ($d = 0.46$ two-tailed).

Figure 1

Participant Flow Through the Study



Note. Analysis sample sizes are patients with a visual acuity change score (average of both eyes) at each wave. Patients considered by the surgeon but not referred were not recorded.

Design

The study was a two-arm, parallel-group randomized controlled trial comparing a fast healing condition (n = 76) with a standard healing (active control) condition (n = 71). Allocation was determined by a randomization sequence generated in R and applied in order of enrollment, with each newly enrolled participant assigned to the condition corresponding to their enrollment number. The surgeon and all clinical staff who conducted the visual acuity examinations were blind to condition, as were participants, who were told only that the study concerned the relationship between psychological factors and recovery following cataract surgery. The study team was not blind to condition, but had no role in outcome assessment.

Materials

Preoperative Information Video

Both conditions watched a short preoperative information video, filmed with the participants' own surgeon, in which he provided basic recovery information following cataract surgery, including expected timeframes for symptoms to resolve and for noticeable improvements in vision to occur. The two videos were identical except for the timeframes given. In the standard healing condition, the surgeon gave the standard timeframes ophthalmologists typically provide to cataract surgery patients: it is normal for vision to be blurry for up to 48 hours; vision often noticeably improves within 48 hours and certainly by 1 week; and itchiness, irritation, and light sensitivity resolve within a couple of weeks. In the fast healing condition, each timeframe was shifted earlier: it is normal for vision to be blurry for 24 to 48 hours; vision often noticeably improves within 24 hours and certainly by 5–7 days; and itchiness, irritation, and light sensitivity resolve within 7 to 14 days (Table 1). The differences between conditions were deliberately subtle in order to present minimal risk to participants, per Harvard's Institutional Review Board. The video was followed by six multiple-choice questions, three of which asked specifically about the healing times provided in the video, in order to reinforce the timeframes presented.

Table 1

Healing Timeframes Presented in the Preoperative Video, by Condition

Video statement	Standard healing	Fast healing
It is normal for vision to be blurry for…	up to 48 hours	24 to 48 hours
Vision often noticeably improves within…	48 hours, certainly by 1 week	24 hours, certainly by 5–7 days
Itchiness, irritation, and light sensitivity resolve within…	a couple of weeks	7 to 14 days

Note. The three healing-time quiz questions that followed each video used the same sentence stems, with the correct answer matching the participant's condition.

Daily Check-In Surveys

Participants in both conditions received 20 daily check-in surveys, 10 after each surgery, sent between the 1-day and 2-week follow-up visits. In the fast healing condition, the surveys were designed to prompt participants to notice small or subtle changes in their healing. On the first day, participants were asked questions such as, "At the present time, how would you describe your eyesight using both eyes (with glasses or contact lenses, if you wear them)?" rated from 1 (*worst possible eyesight*) through 5 (*some blurriness but okay*) to 10 (*best possible eyesight*). From the second day onward, each item was prefaced with "compared to yesterday" and the scale was re-anchored accordingly (e.g., 1 = *much better eyesight than yesterday*, 5 = *about the same as yesterday*, 10 = *much worse eyesight than yesterday*). Six items were asked each day, covering overall feelings of health, eyesight using both eyes, pain or discomfort in and around the eyes, difficulty with daily activities such as reading and cooking, feeling limited by one's vision, and how frequently the participant was intentionally spending a few minutes paying attention to small changes in their vision. This design was meant to nudge participants to notice small changes in their recovery, since we believe attending to subtle changes in physiology shapes how physiological processes unfold, analogous to paying attention to the breath and in so doing changing how the respiratory process unfolds (Aungle, Matta, Loecher, et al., 2026). In previous work, Langer has argued that attending to symptom variability may be the key to understanding the placebo effect (Langer, 2023; Pagnini et al., 2015; Zilcha-Mano & Langer, 2016), which is especially relevant to chronic health conditions in which people assume their symptoms are stable or are inevitably worsening (e.g., Leventhal et al., 2016).

In the standard healing condition, the surveys were matched in frequency and approximate length, but the items asked about participants' current state rather than change, and attention was not directed to the physiology of healing. Items included free-response questions such as "What were you doing right before this survey?" and "How have you been feeling today? We are interested in the levels of stress, anxiety, and pain you have been experiencing," a question about the amount and quality of rest, and Likert-type items such as "I am feeling happy right now" and "I am feeling stressed right now," rated from 1 (*strongly disagree*) to 6 (*strongly agree*).

Enrollment and Final Surveys

The enrollment survey collected demographic information and several psychosocial variables known to influence physical healing, as in our prior study of perceived time and healing (Aungle & Langer, 2023): chronic stress using the 14-item Perceived Stress Scale (PSS-14; Cohen et al., 1983), anxiety using the anxiety subscale of the Hospital Anxiety and

Depression Scale (HADS-A; Zigmond & Snaith, 1983), and mindfulness using the 14-item Langer Mindfulness Scale (LMS-14; Pirson et al., 2012). Self-reported visual function was assessed with the National Eye Institute Visual Function Questionnaire (NEI-VFQ-25; Mangione et al., 1998). We also measured General Healing Expectations (GHE) – the extent to which participants believe they heal faster than other people – with three items ("Compared to other people, my recovery time after an injury is generally…") rated on 6-point bipolar scales (*Slower–Faster*, *Harder–Easier*, *More Painful–Less Painful*), since we assume beliefs about physiological processes influence the way they unfold (Aungle, Matta, Chen, et al., 2026). Following Aungle and Langer (2023), we also administered the Ten-Item Personality Inventory (Gosling et al., 2003) to support the cover story. The final study survey, sent three months after the first surgery, repeated the enrollment survey; its primary purpose was to assess change in NEI-VFQ-25 scores from baseline.

Visual Acuity

Objective visual acuity was assessed by the clinical team at the ophthalmologist's office at baseline (prior to the first surgery), 1 day and 1 to 2 weeks after each surgery, and 3 months after the second surgery, using standard clinical Snellen chart examinations of best-corrected visual acuity in each eye. The clinical team also recorded any surgical complications. Snellen values were converted to the logarithm of the minimum angle of resolution (logMAR) prior to analysis. logMAR provides an approximately interval-scaled measure of visual acuity in which equal changes reflect equal proportional changes in visual resolution, making it well suited to modeling change over time (e.g., Holladay, 1997).

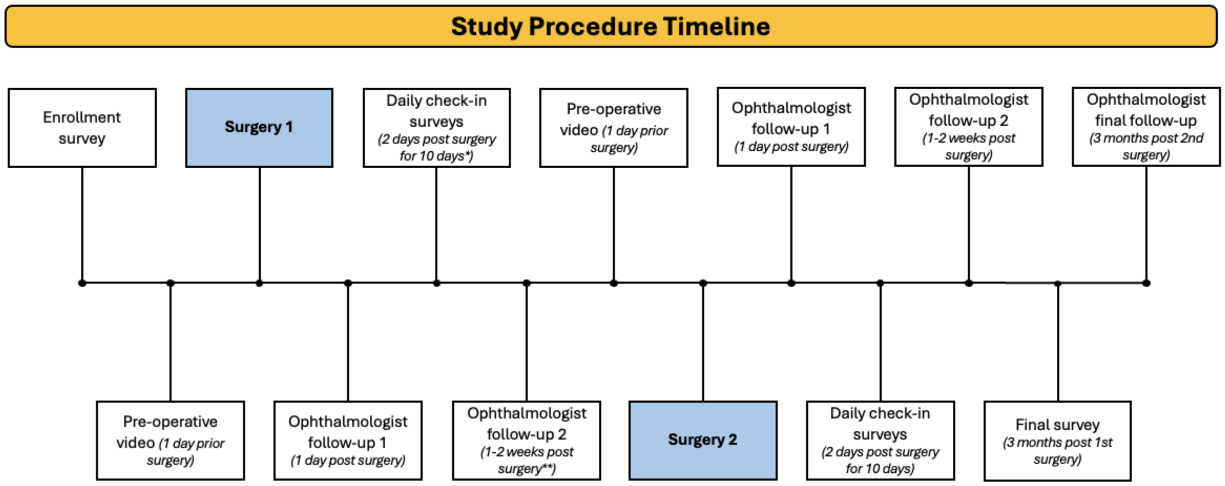
Procedure

Eligible patients were told about the study by their ophthalmologist and, if interested, had their contact information passed to the study team, who sent a brief introductory email with a link to the enrollment survey. Completing the enrollment survey constituted informed consent and formally enrolled participants in the study. Participants were then randomized to condition. Figure 2 summarizes the timeline that followed, which was identical for the two conditions except for the content of the preoperative video and the daily check-in surveys.

One day before the first surgery, the study team sent a link to the preoperative information video; a team member telephoned participants who had not yet watched it. One day after surgery, participants returned to the ophthalmologist's office for a follow-up visit that included a clinical visual acuity examination (Wave 1). Beginning two days after surgery, participants received daily check-in surveys. When the second surgery was scheduled at least two weeks after the first, participants received 10 daily surveys; when the interval was shorter, surveys began 48 hours after the first surgery and ended 48 hours before the second. Approximately two weeks after the first surgery, participants returned for a follow-up visit including a visual acuity examination (Wave 2); for participants whose second surgery fell within two weeks of the first, this examination was conducted on the day of the second surgery, before the procedure. The same sequence – video the day before, examination the day after, 10 daily check-in surveys, and an examination at approximately two weeks – was repeated for the second surgery. Three months after the first surgery, participants completed the final study survey, and three months after the second surgery they returned for a final visual acuity examination (Wave 3). Participation thus lasted approximately three to four months from the first surgery.

Figure 2

Study Procedure Timeline



Note. Both conditions went through the same procedure; only the healing times given in the preoperative information video and the questions asked in the daily check-in surveys differed. *Where the second surgery was less than 2 weeks after the first, check-in surveys began 48 hours after the first surgery and ended 48 hours before the second, and **the follow-up examination occurred on the date of the second surgery (before the procedure).

Participants received a total of $25 in Amazon gift cards for completing all parts of the study: $5 for watching each preoperative video and answering the accompanying questions ($10 total), $5 for completing each round of 10 daily check-in surveys ($10 total), and $5 for completing the final study survey.

Analytic Strategy

Outcome Variables

Rate of healing was operationalized as the change in logMAR visual acuity from baseline to the 1-day (Wave 1) and 1-to-2-week (Wave 2) postoperative examinations, averaged across both eyes. Negative change scores indicate improvement. Overall surgical outcome was operationalized as the change in logMAR from baseline to the 3-month examination (Wave 3) and the change in NEI-VFQ-25 composite score from enrollment to the final survey.

Predictors

Daily attention to healing. For participants in the fast healing condition, we computed a single daily attention score from the six check-in items. Each response was rescored as the difference between 5 (the “same as yesterday” anchor) and the selected rating, so that positive values indicate noticing improvement and negative values indicate noticing worsening, and these values were averaged across all six items and all completed surveys. The score is thus a composite measure of attention to healing variability throughout the recovery period. Participants in the standard healing condition, whose surveys did not direct attention to healing, were assigned a daily attention score of zero by design.

Sensitivity to the video. Sensitivity to the healing-time information was operationalized as the mean number of the three healing-time quiz questions answered correctly across the two viewings (range 0 to 3). Participants who answered more of these questions correctly are assumed to have more fully absorbed the expected healing times conveyed in the video. Quiz data were available for 73 treatment and 68 control participants; for the remainder, sensitivity was imputed at the grand mean. Daily attention scores were missing for 3 treatment participants who did not complete any check-in surveys; these participants are excluded from the attention models.

Covariates. All models adjust for age and the baseline psychosocial measures (PSS-14, HADS-A, LMS-14), mean-centered.

Statistical Models

Whether the intervention and attention to healing predict recovery at any single time point is less central to our reasoning than whether these effects are jointly present across the recovery period. Because the same participants contribute outcomes at multiple postoperative time points, we use seemingly unrelated regression (SUR) as our primary analytic framework. SUR estimates separate regression equations for each wave simultaneously – one predicting Wave 1 logMAR change and another predicting Wave 2 logMAR change – using the same set of predictors. Critically, SUR models the correlation between residuals across equations, reflecting the fact that a patient who heals unexpectedly well at one time point is likely to heal unexpectedly well at another. This allows a joint chi-squared test of whether a given predictor is simultaneously zero across waves, a principled way to evaluate whether an effect is present across the recovery period while correctly accounting for the non-independence of repeated observations.

When predictors are identical across equations, as here, the SUR point estimates for each wave are equivalent to wave-by-wave ordinary least squares (OLS); the value of SUR lies in the joint test. We complement the SUR analyses with pooled OLS, in which observations are stacked across waves with a wave fixed effect and a single regression is estimated. Pooled OLS yields a single summary coefficient and confidence interval that is useful for interpretation, but it treats observations as independent, which inflates the effective sample size and can produce overly liberal *p*-values. We therefore report the two together: pooled OLS for the point estimate and interval, and the SUR joint test as the more conservative test of significance. Because our predictions are directional, we report one-tailed tests unless otherwise indicated (Hales, 2024).

Because daily attention was measured only in the fast healing condition, the models above can examine heterogeneity only within that condition. We therefore supplement them with two analyses that use the full sample. First, we cross condition with General Healing Expectations (median split) to form four cells, compute covariate-adjusted cell means for all 13 outcomes, and use a permutation test (5,000 joint shuffles of the cell labels) to evaluate whether the predicted cell – fast healing with high GHE – is the best-recovery cell more often than chance would produce. Second, we split fast healing participants by belief (sensitivity or GHE) and daily attention and compare each subgroup's mean change in visual acuity at each wave with that of control participants at the same belief level.

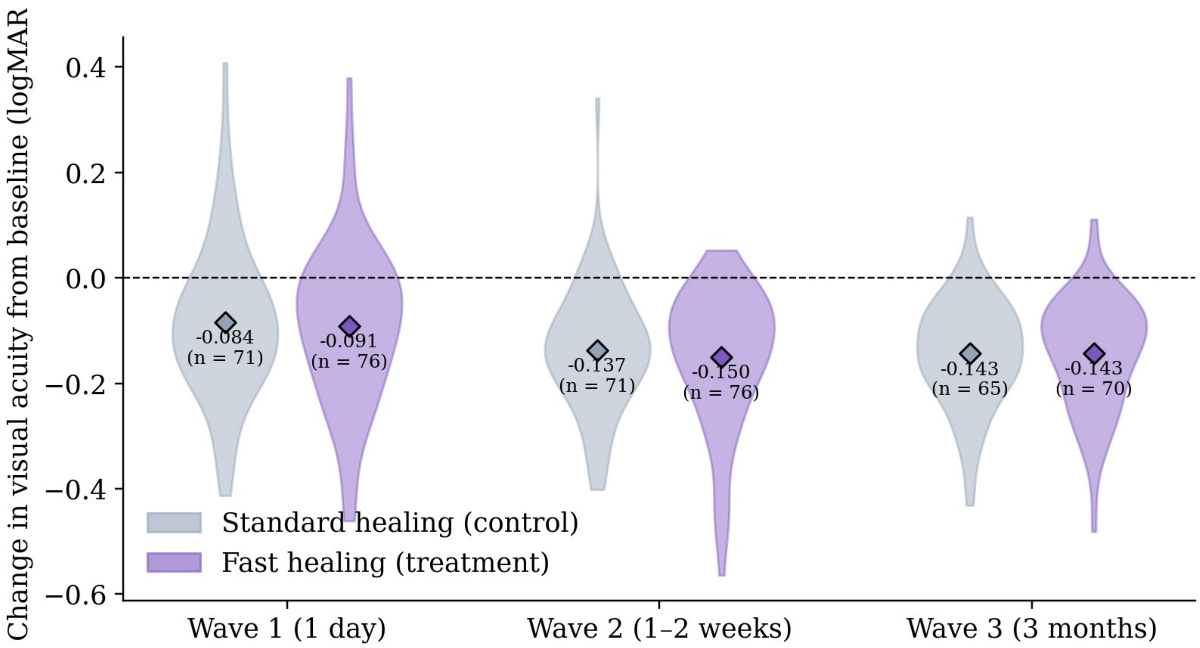
Results

Descriptive Results

Figure 3 presents violin plots comparing the distribution of change in visual acuity (logMAR) between the fast healing (treatment) and standard healing (control) groups at each wave, averaged across both eyes. Both groups improved from baseline at every time point, with mean change scores consistently below zero. The treatment group showed slightly larger mean improvements at Waves 1 and 2: at Wave 1, the mean change was −0.091 logMAR (n = 76) for the treatment group compared to −0.084 (n = 71) for the control group, and at Wave 2, −0.150 (n = 76) versus −0.137 (n = 71). By Wave 3, the group averages converged (−0.143, n = 70, versus −0.143, n = 65), partly due to attrition and partly due to more similar outcomes once fully healed. Although these group-level differences are modest, they do not account for the two prongs of our intervention: internalizing the faster expected healing times provided in the preoperative video and attending to subtle changes in healing throughout the recovery process.

Figure 3

Visual acuity change from baseline



Note. Violin plots of change in visual acuity (logMAR) from preoperative baseline, averaged across both eyes, by condition and wave. Diamonds = group means (values and *n* shown). Negative values indicate better postoperative vision; dashed line = no change from baseline.

Hypothesis testing

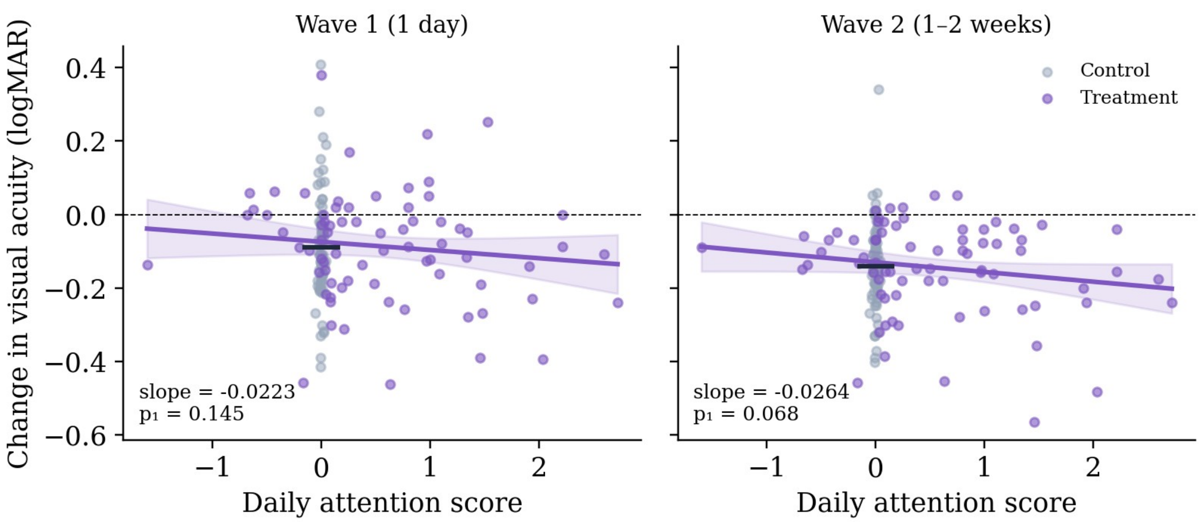
Hypothesis 1

To test our first hypothesis, we regressed logMAR change at each wave on the daily attention score (see Method), adjusting for age, PSS-14, LMS-14, and HADS-A.

Figure 4 plots the relationship between daily attention to healing and visual acuity recovery at each wave, with the solid purple line showing the treatment group slope and the short gray line indicating the control group mean. The regressions are adjusted for age, PSS-14, LMS-14, and HADS-A at enrollment. At Wave 1, the slope of attention to healing on change in visual acuity was −0.0223 (p_1 = 0.145); at Wave 2, the slope was −0.0264 (p_1 = 0.068). The negative slopes at Waves 1 and 2 indicate that greater daily attention to healing variability was associated with larger improvements in visual acuity (i.e., more negative logMAR change scores), directionally consistent with our prediction, but the relationship was not significant at conventional levels at either wave. The effect was strongest at Wave 2, which corresponds to the 1–2 week follow-up visits – the time point at which the cumulative influence of daily attention would be expected to be most apparent.

Figure 4

Acuity recovery vs. daily attention score

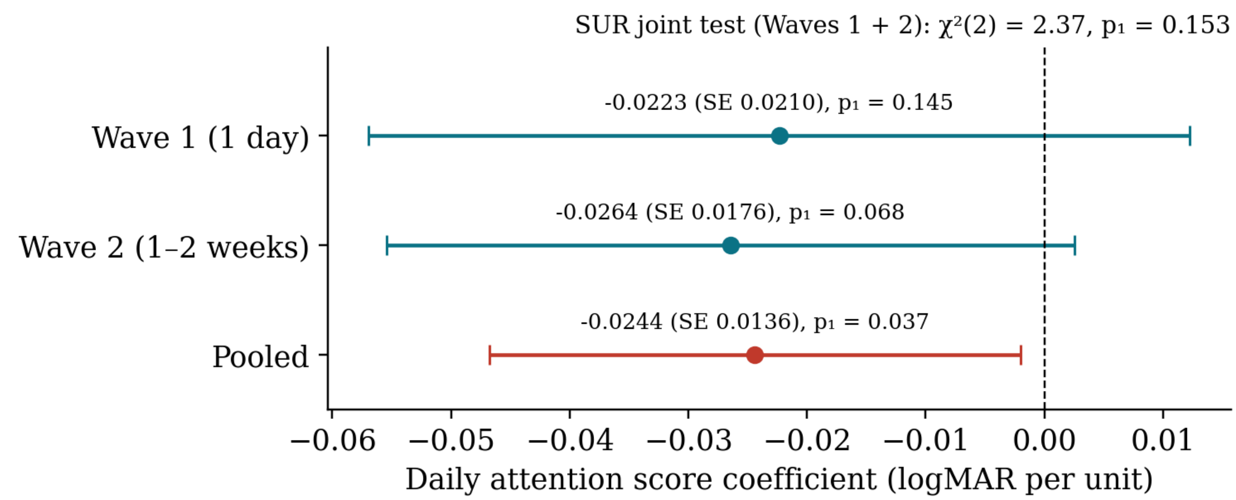


Note. Scatter plots of daily attention to healing variability against visual acuity change at Wave 1 (1 day) and Wave 2 (1–2 weeks). Purple line = model-predicted values for the treatment group at covariate means (age, HADS-A, PSS-14, LMS-14), with 90% confidence band; short dark line = control group mean. Controls cluster at $x = 0$ because their surveys did not direct attention to healing. p-values one-tailed. $N = 144$.

Because the daily attention coefficient is estimated at both Wave 1 and Wave 2, the SUR joint test asks whether attention to healing is associated with recovery across the two waves taken together (Figure 5). The wave-level coefficients were both in the predicted direction, but the joint test did not reach significance ($\chi^2(2) = 2.37$, $p_1 = 0.153$), indicating that while the pattern of effects was consistent with our prediction, the relationship between attention to healing variability and actual healing was not especially strong on its own. Attention to variability was nonetheless directionally associated with healing in the manner the Embodied Models of Health framework would predict, and attention is only one component of that model. The beliefs and expectations component needs to be included to evaluate the EMH predictions more fully.

Figure 5

Daily attention score coefficient by wave



Note. Coefficient on the daily attention score at each wave, with 90% confidence intervals and one-tailed *p*-values; the pooled estimate (stacked OLS with wave fixed effect, $N = 288$ observations) is shown for reference only, since it does not account for the correlation of errors across waves. Controls: age, HADS-A, PSS-14, LMS-14. $N = 144$ patients at each wave.

Table 2

Daily Attention Score: Full Specification (Wave 1, Wave 2, Pooled)

	Wave 1 (1 day) (1)	Wave 2 (1–2 weeks) (2)	Pooled (3)
Treatment group	0.016	0.012	0.014
	(0.027)	(0.023)	(0.017)
Daily Attention Score	−0.022	−0.026*	−0.024**
	(0.021)	(0.018)	(0.014)
Age	0.006***	0.004***	0.005***
	(0.002)	(0.002)	(0.001)
Anxiety (HADS-A)	−0.031	−0.025	−0.028*
	(0.033)	(0.027)	(0.021)
Stress (PSS-14)	0.029	0.025	0.027*
	(0.030)	(0.025)	(0.019)
Mindfulness (LMS-14)	−0.004	−0.003	−0.003
	(0.017)	(0.014)	(0.011)
Wave 2 (FE)			−0.055***
			(0.016)
Constant	−0.090***	−0.142***	−0.088***
	(0.017)	(0.015)	(0.014)
Observations	144	144	288
R^2	0.095	0.076	0.120
Adjusted R^2	0.055	0.035	0.098

Note. Dependent variable: change in logMAR visual acuity from baseline, averaged across both eyes. Standard errors in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$ (one-tailed). All covariates mean-centered. Daily Attention Score = 0 for controls by design. Pooled: stacked OLS with wave fixed effect; N = stacked observations. SUR joint test of the attention coefficient (Waves 1 + 2): $\chi^2(2) = 2.37$, $p_1 = 0.153$.

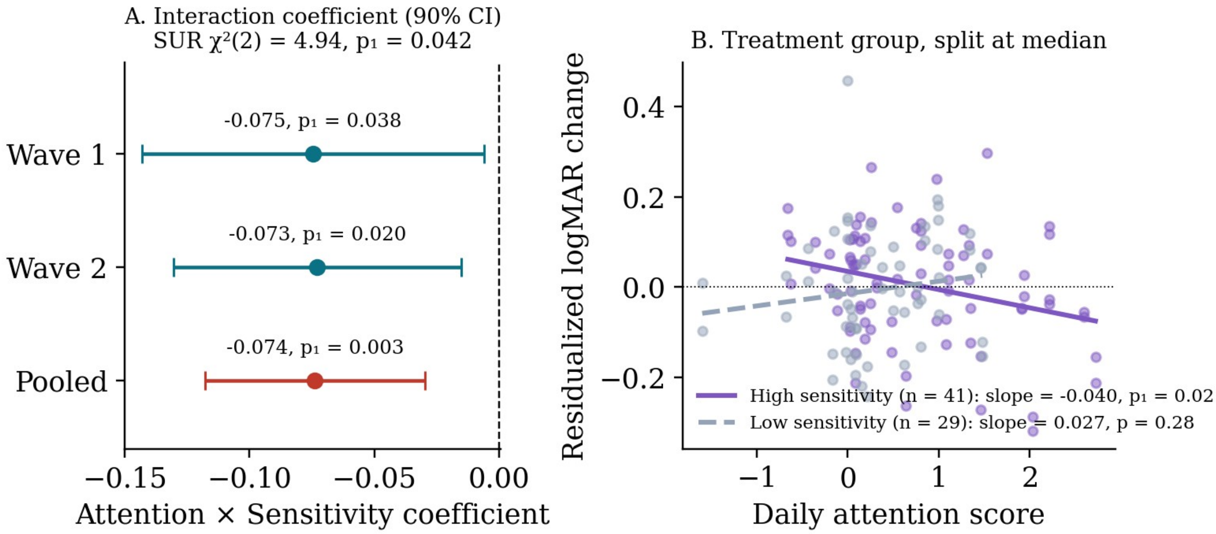
Hypothesis 2

To test our second hypothesis, we examined whether the effect of daily attention to healing on visual acuity recovery was moderated by participants' sensitivity to the healing-time information provided in the preoperative video. As described in the Hypotheses section, sensitivity was operationalized as the mean number of healing-time questions answered correctly in the survey that followed each preoperative video viewing. Participants who answered more of these questions correctly are assumed to have more fully absorbed the expected healing times conveyed in the video, which, according to our Embodied Models of Health framework, should amplify the effect of attending to healing variability.

We included an Attention × Sensitivity interaction term in our SUR and pooled OLS models, controlling for age, PSS-14, LMS-14, and HADS-A at enrollment. The results supported our prediction. As shown in Panel A of Figure 6, the Attention × Sensitivity interaction coefficient was negative at both waves (Wave 1: −0.075, $p_1 = 0.038$; Wave 2: −0.073, $p_1 = 0.020$) and in the pooled model (−0.074, *SE* = 0.027, $p_1 = 0.003$), with the 90% confidence interval of the pooled estimate falling entirely below zero, indicating that participants who both attended to healing variability and internalized the video's healing times experienced larger improvements in visual acuity. The SUR joint test across Waves 1 and 2 was significant ($\chi^2(2) = 4.94$, $p_1 = 0.042$), indicating the interaction effect was jointly present across the recovery period (see full results in Table 3).

Figure 6

Daily Attention × Sensitivity interaction



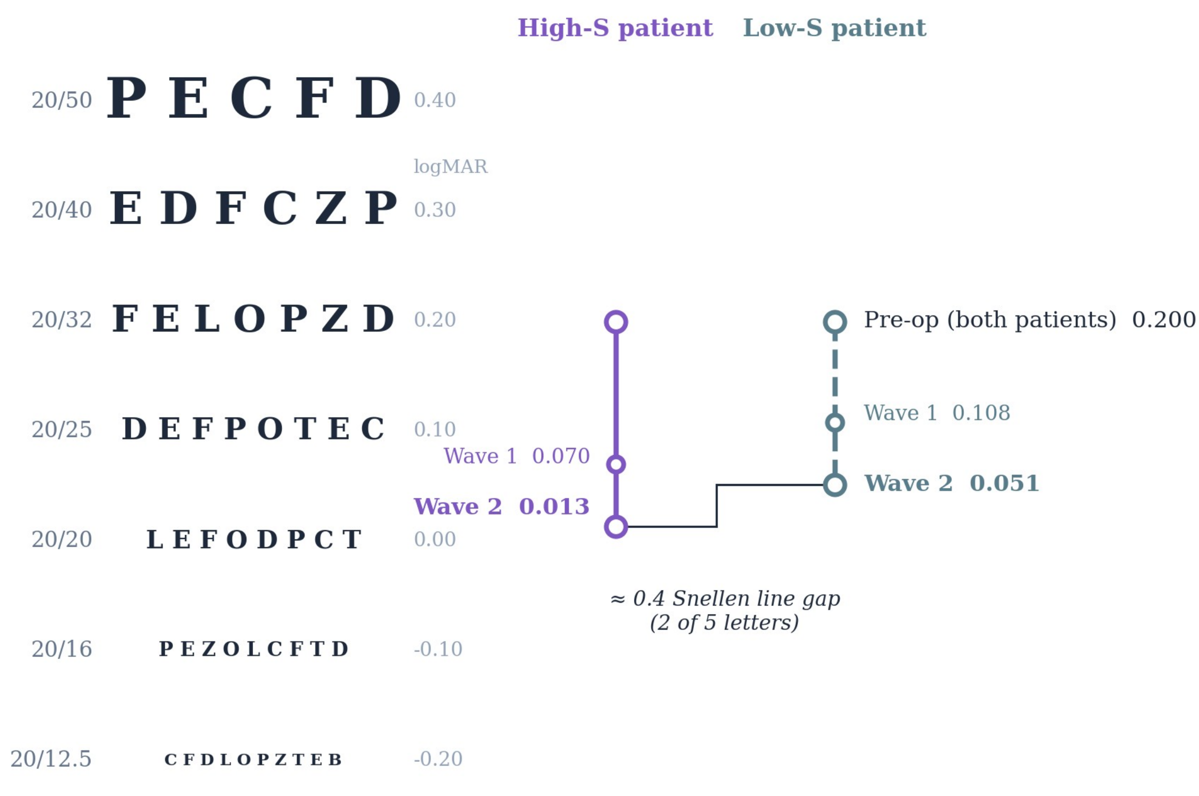
Note. Panel A: Attention × Sensitivity interaction coefficient at each wave and pooled, with 90% confidence intervals (one-tailed). Panel B: treatment group only, split at median sensitivity (≥ 2.0, n = 41; < 2.0, n = 29); points are residuals of pooled Wave 1 and Wave 2 logMAR change net of covariates and wave; lines are within-subgroup slopes with standard errors clustered by patient.

To put the Attention × Sensitivity coefficient in perspective, take two patients with identical preoperative visual acuity who undergo the same surgery, and increase each patient's daily attention to healing by one standard deviation. The model predicts that the patient one standard deviation higher in sensitivity to the video ends up 0.038 logMAR better by Wave 2 than the patient one standard deviation lower — roughly 0.4 of a Snellen line, or 2 of the 5 letters on the next line down: a meaningful, clinically visible gap. Figure 7 shows what this looks like on a Snellen chart. Both patients begin at 20/32 (0.20 logMAR, near the treatment-group mean baseline of 0.18); the Wave 2 positions are predicted by the Attention × Sensitivity model at +1

SD daily attention with covariates at their means, and the vertical separation between the two arrows is the gap attributable to the interaction.

Figure 7

Two Patients, Same Surgery: Predicted Wave 2 Acuity by Video Sensitivity



Note. Panel A: model-predicted visual acuity (logMAR; lower = better) for the high- and low-sensitivity groups in the fast healing condition at Wave 1 and Wave 2, from the Attention × Sensitivity model with daily attention set to +1 *SD* (1.37) and covariates at their means. Each group begins at the treatment group's own observed mean preoperative acuity. Letter rows are illustrative; 0.10 logMAR corresponds to one Snellen line. Panel B: Attention × Sensitivity interaction coefficient at each wave with 90% confidence intervals and one-tailed *p*-values.

And this gap could widen further as the high-sensitivity patient continues to notice more variability in their healing.

Panel B of Figure 6 illustrates this interaction by splitting the treatment group at the median sensitivity score (range = 0 to 3, median = 2.0) and regressing residualized logMAR change (net of the covariates and wave) on daily attention within each subgroup, with standard errors clustered by patient. Among high-sensitivity participants (those who answered at least two of the three healing-time questions correctly on average; *n* = 41), the relationship between daily attention to healing and visual acuity improvement was negative and statistically significant (slope = −0.040, $p_1 = 0.021$): the more they attended to variability in their healing process, the more their vision improved. Among low-sensitivity participants (*n* = 29), there was no such relationship (slope = +0.027, *p* = 0.28, two-tailed). In other words, attending to healing variability predicted faster recovery for participants who had internalized the faster healing times conveyed in the preoperative video. This pattern is consistent with our EMH theoretical framework, which predicts that attention to physiological processes amplifies healing in a direction consistent with one's expectations. Participants who internalized the expectation of faster healing and then noticed day-to-day changes consistent with that expectation healed more than those who did not. Importantly, simply attending to variability alone did not have any consistent effect: low-sensitivity participants, if anything, seemed to experience slightly worse outcomes the more they claimed to notice improvement.

Table 3

A×S Interaction: Full Specification (Wave 1, Wave 2, Pooled)

	Wave 1 (1 day) (1)	Wave 2 (1–2 weeks) (2)	Pooled (3)
Treatment group	0.037*	0.031*	0.034**
	(0.028)	(0.024)	(0.018)
Sensitivity (S)	0.056**	0.029*	0.042***
	(0.025)	(0.021)	(0.016)
Treatment × Sensitivity	−0.020	0.031	0.006
	(0.042)	(0.035)	(0.027)
Daily Attention Score (A)	−0.032*	−0.037**	−0.034***
	(0.021)	(0.018)	(0.014)
Attention × Sensitivity (A×S)	−0.075**	−0.073**	−0.074***
	(0.042)	(0.035)	(0.027)
Age	0.006***	0.003**	0.004***
	(0.002)	(0.002)	(0.001)
Anxiety (HADS-A)	−0.046*	−0.036*	−0.041**
	(0.033)	(0.027)	(0.021)
Stress (PSS-14)	0.035	0.027	0.031*
	(0.029)	(0.024)	(0.019)
Mindfulness (LMS-14)	−0.008	−0.006	−0.007
	(0.017)	(0.014)	(0.011)
Wave 2 (FE)			−0.055***
			(0.015)
Constant	−0.104***	−0.148***	−0.099***
	(0.018)	(0.015)	(0.014)
Observations	144	144	288
R^2	0.147	0.123	0.164
Adjusted R^2	0.089	0.064	0.134

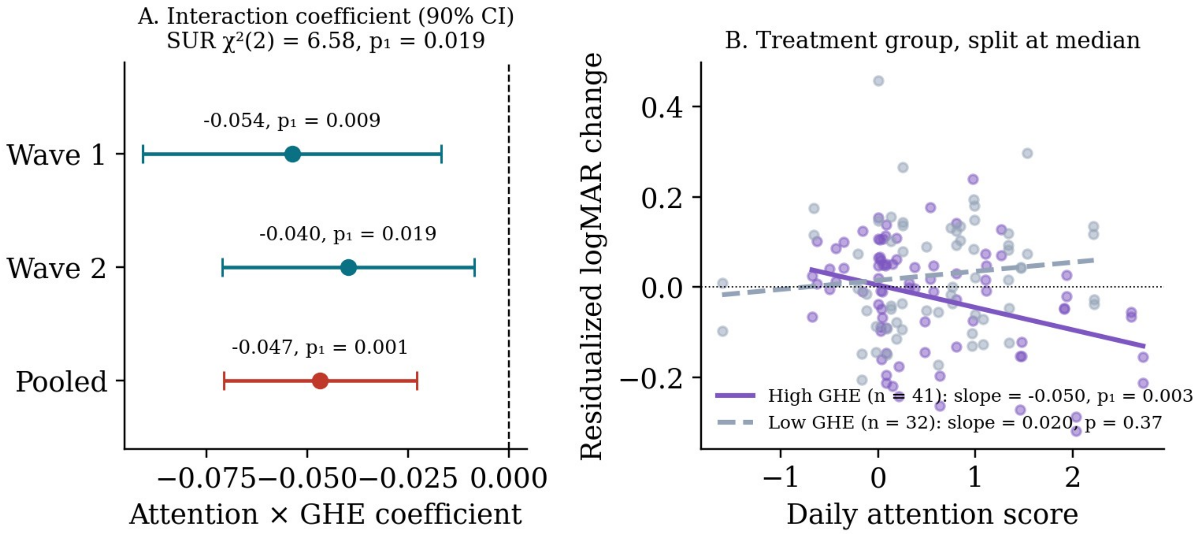
Note. Dependent variable: change in logMAR visual acuity from baseline, averaged across both eyes. Standard errors in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$ (one-tailed). All continuous covariates mean-centered. Sensitivity mean-centered and imputed to the grand mean (0) for participants without quiz data. Key coefficient: A×S. Pooled: stacked OLS with wave fixed effect; *N* = stacked observations. SUR joint test of the A×S coefficient (Waves 1 + 2): $\chi^2(2) = 4.94$, $p_1 = 0.042$.

As a robustness check on this interaction, we estimated the same model reported for Hypothesis 2 with the General Healing Expectations measure collected at enrollment in place of the sensitivity variable. We measured General Healing Expectations in Aungle and Langer (2023) as well and found a similar beliefs × attention interaction effect on healing (Aungle, Matta, Loecher, et al., 2026). The results almost perfectly parallel those for Hypothesis 2.

Figure 8 presents the results. Panel A shows the OLS coefficients for the Attention × General Healing Expectations interaction on visual acuity change at each wave, as well as the pooled estimate. The interaction coefficient was significant at both Wave 1 (−0.054, $p_1 = 0.009$) and Wave 2 (−0.040, $p_1 = 0.019$), and highly significant in the pooled model (−0.047, *SE* = 0.015, $p_1 < .001$). The SUR joint test across Waves 1 and 2 confirmed joint significance ($\chi^2(2) = 6.58$, $p_1 = 0.019$, *N* = 144 patients). Panel B illustrates this interaction by splitting the treatment group at the median General Healing Expectations score (raw = 4.67). Among high-GHE participants (*n* = 41), greater daily attention to healing strongly predicted larger improvements in visual acuity (slope = −0.050, $p_1 = 0.003$). Among low-GHE participants (*n* = 32), the relationship was non-significant (slope = 0.020, *p* = 0.38, two-tailed). This pattern closely parallels the Attention × Sensitivity interaction reported for Hypothesis 2, with a somewhat stronger effect: the SUR joint test was more clearly significant here ($p_1 = 0.019$) than for the sensitivity variable ($p_1 = 0.042$), and the high-GHE slope (−0.050) was steeper than the high-sensitivity slope (−0.040).

Figure 8

Attention × General Healing Expectations interaction on healing (Wave 1 & Wave 2 logMAR)



Note. Panel A: Attention × GHE interaction coefficient at each wave and pooled, with 90% confidence intervals (one-tailed), and SUR joint test. Panel B: treatment group only, split at median GHE (n = 41 high, 32 low); points are residuals of pooled Wave 1 and Wave 2 logMAR change net of covariates and wave; lines are within-subgroup slopes with standard errors clustered by patient.

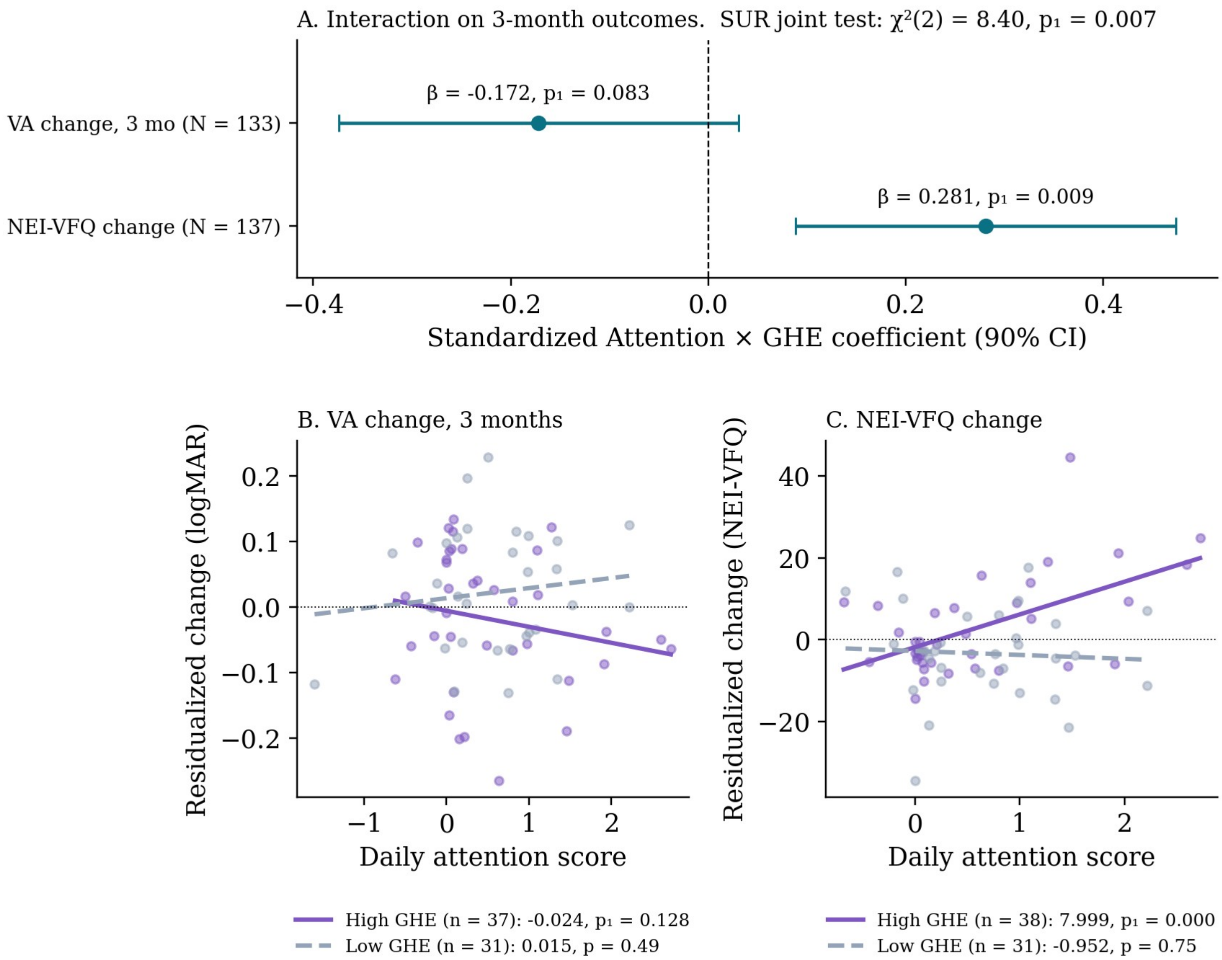
Additional Analyses

Our medical team collaborators cautioned us at the outset that even if our intervention accelerates healing, we should not expect much of a difference in outcomes (since most patients end up with 20/20 vision once fully healed). Nonetheless, we collected outcome measures of visual acuity out of curiosity. We thought there was a reasonable chance the intervention would generalize beyond healing to overall outcomes. The change from baseline to 3-month postoperative visual acuity and from NEI-VFQ scores at enrollment to NEI-VFQ scores 3 months after surgery served as our measures of overall outcomes.

Figure 9 presents the results. Panel A shows standardized OLS coefficients for the Attention × General Healing Expectations interaction on two outcome measures: change in visual acuity from baseline to 3 months (*N* = 133) and change in vision-related quality of life as measured by the NEI-VFQ-25 composite (*N* = 137). For visual acuity change at 3 months, the standardized interaction coefficient was negative (indicating greater improvement) but did not reach significance ($\beta = -0.172$, $p_1 = 0.083$). For vision-related quality of life, the interaction coefficient was positive (indicating greater improvement) and significant ($\beta = 0.281$, $p_1 = 0.009$). Panels B and C illustrate these interactions by splitting the treatment group at the median General Healing Expectations score. Among high-GHE participants, greater daily attention to healing was associated with larger improvements in both visual acuity (slope = −0.024, $p_1 = 0.128$; *n* = 37) and vision-related quality of life (slope = 8.00, $p_1 < .001$; *n* = 38). We report the relationship between GHE and attention to healing for both objective changes in logMAR visual acuity and changes in NEI-VFQ scores because the effect is jointly significant across these measures (SUR joint test $\chi^2(2) = 8.40$, $p_1 = 0.007$, *N* = 126). Among low-GHE participants, attention to healing showed no meaningful relationship with either outcome (visual acuity: slope = 0.015, *p* = 0.49; NEI-VFQ: slope = −0.95, *p* = 0.76; two-tailed).

Figure 9

Attention × General Healing Expectations interaction on long-term outcomes (3 months)

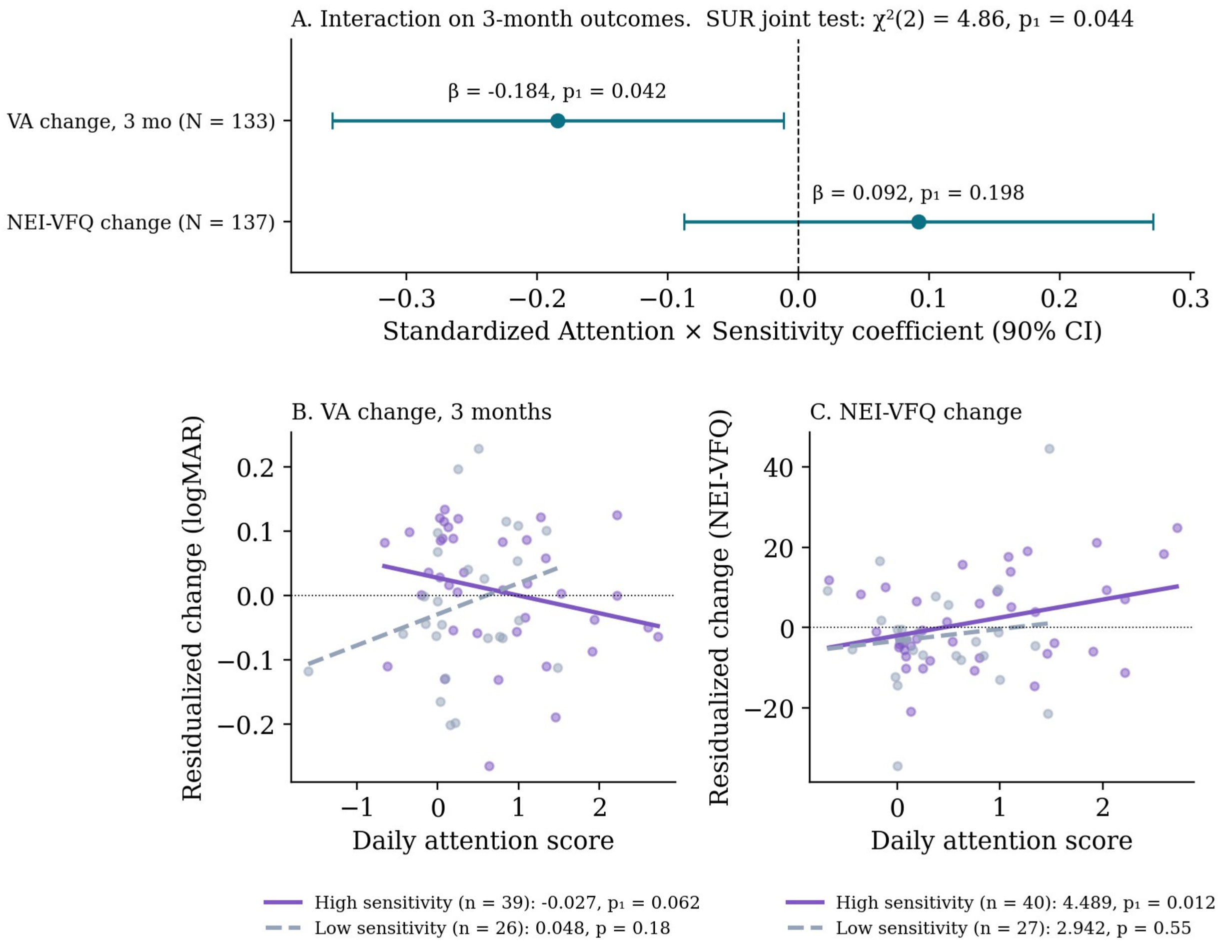


Note. Panel A: standardized OLS coefficients for the Attention × GHE interaction with 90% confidence intervals (one-tailed). Panels B and C: treatment group only, split at the median GHE score (raw = 4.67); points are residuals net of covariates.

As a robustness check on these outcome analyses, we re-estimated the same models with Sensitivity in place of General Healing Expectations (Figure 10). The Attention × Sensitivity interaction on 3-month visual acuity change was significant ($\beta = -0.184$, $p_1 = 0.042$; $N = 133$), somewhat stronger than the corresponding A×GHE estimate, while the interaction on NEI-VFQ change was in the predicted direction but weaker ($\beta = 0.092$, $p_1 = 0.198$; $N = 137$); the SUR joint test across the two outcomes was significant ($\chi^2(2) = 4.86$, $p_1 = 0.044$, $N = 126$). Among high-sensitivity participants, greater daily attention predicted larger improvements in both visual acuity (slope = −0.027, $p_1 = 0.062$; $n = 39$) and NEI-VFQ scores (slope = 4.49, $p_1 = 0.012$; $n = 40$), whereas neither relationship held among low-sensitivity participants. Sensitivity and General Healing Expectations are essentially uncorrelated in our sample ($r = -0.02$), so the two sets of results constitute independent evidence for the belief × attention interaction the EMH framework postulates. We would not expect the two moderators to be equally relevant to every outcome: sensitivity indexes beliefs about how quickly healing unfolds in the days immediately following cataract surgery, not beliefs about eventual outcomes, so it is unlikely to be the most relevant belief for 3-month quality of life. We report it here because sensitivity is tied directly to the manipulation: the preoperative video was the component of the intervention we controlled experimentally, and sensitivity indexes how fully each participant absorbed the timeline it conveyed, whereas General Healing Expectations were measured at enrollment and never manipulated.

Figure 10

Attention × Sensitivity interaction on long-term outcomes (3 months)



Note. Panel A: standardized OLS coefficients for the Attention × Sensitivity interaction with 90% confidence intervals (one-tailed). Panels B and C: treatment group only, split at the median sensitivity score (≥ 2.0 vs. < 2.0); points are residuals net of covariates (age, HADS-A, PSS-14, LMS-14). Sensitivity is imputed at the grand mean for participants without quiz data.

This pattern suggests the intervention did generalize to some extent, which, from a theoretical perspective, makes sense: cultivating constructive beliefs and attending to variability consistent with those beliefs is a more general process.

Testing the Belief × Attention Prediction Against Active Controls

The analyses reported so far can only examine heterogeneity within the fast healing condition, because daily attention to healing was, by design, measured only for participants whose surveys directed attention to healing. We therefore conducted two further analyses that use the full sample, including the active controls, to test the same belief × attention prediction from a different angle.

Permutation Test Across All Outcomes

The EMH framework makes a simple prediction about the design as a whole: participants who received the fast healing intervention *and* held favorable beliefs about their own healing should show the best recovery. We therefore crossed condition with General Healing Expectations (split at the median, 4.67) to form four cells and, for each of our 13 outcomes – visual acuity change at Waves 1, 2, and 3 and the 3-month change in the NEI-VFQ composite and its nine subscales – computed covariate-adjusted cell means (OLS with age, HADS-A, PSS-14, and LMS-14 at their grand means). Table 4 reports the adjusted means. The predicted cell, Fast × High GHE, showed the best recovery on 11 of the 13 outcomes (2 of 3 visual acuity waves and 9 of 10 NEI-VFQ measures). To evaluate how unusual this is, we ran a permutation test in which the condition × GHE cell labels were shuffled across participants 5,000 times; each permutation reshuffled the labels once and recomputed all 13 outcomes from that same shuffle, so the correlation among outcomes is preserved rather than assumed away. Under the null, the predicted cell is expected to be best on 3.1 of 13 outcomes; the observed count of 11 occurred in 1.5% of permutations (p = 0.015).

Table 4

Covariate-Adjusted Change From Baseline by Condition × General Healing Expectations Cell

Outcome	Fast × High GHE	Standard × High GHE	Fast × Low GHE	Standard × Low GHE
Wave 1 (1 day)	−0.102	**−0.103**	−0.064	−0.079
Wave 2 (1–2 weeks)	**−0.160**	−0.152	−0.126	−0.133
Wave 3 (3 months)	**−0.158**	−0.145	−0.122	−0.143
NEI-VFQ composite	**+16.2**	+10.8	+9.0	+11.9
General vision	**+28.4**	+24.0	+19.8	+22.8
Near activities	**+25.4**	+13.2	+13.4	+14.1
Distance activities	**+23.1**	+15.0	+15.3	+18.4
Role difficulties	**+21.3**	+8.4	+13.7	+15.9
Dependency	**+9.4**	+7.8	+0.8	+6.5
Mental health	**+20.7**	+13.8	+8.5	+14.0
Ocular pain	**+6.8**	-0.8	+5.3	+4.0
Peripheral vision	**+13.4**	+12.3	+6.8	+12.1
Social functioning	+6.7	**+7.9**	+3.5	+4.5

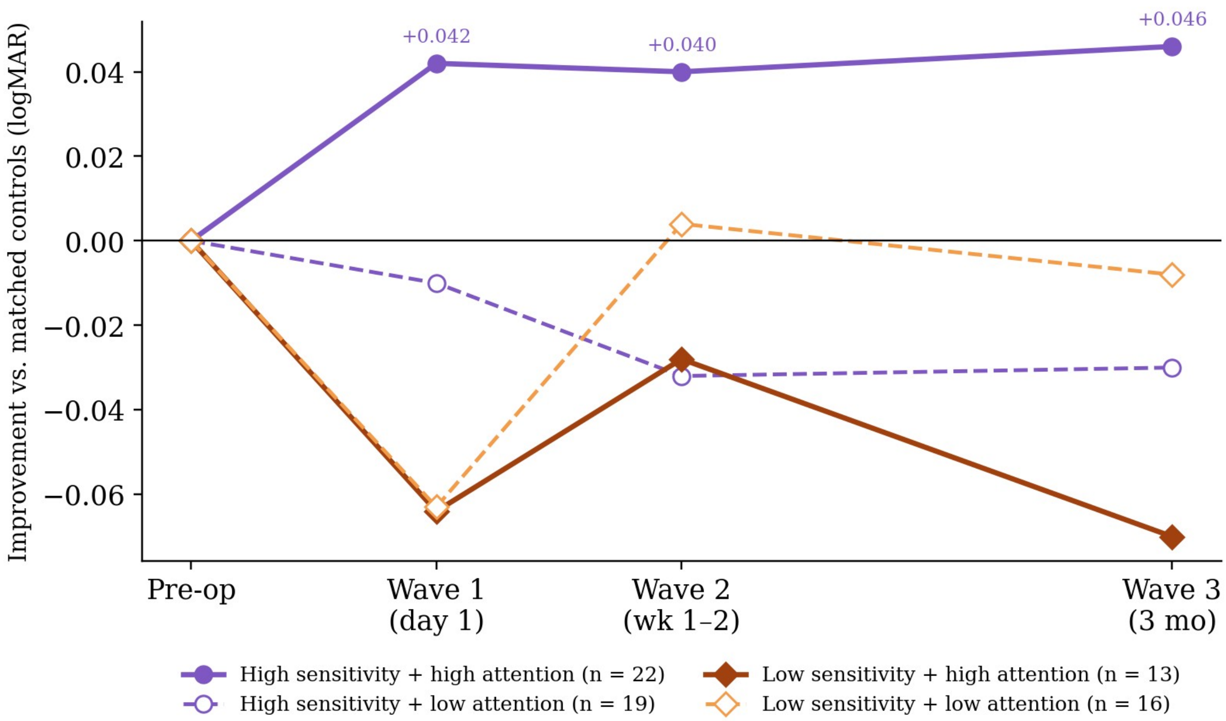
Note. Visual acuity rows are change in logMAR (negative = improvement); NEI-VFQ rows are change on 0–100 scales (positive = improvement). Means are OLS-adjusted for age, HADS-A, PSS-14, and LMS-14 at their grand means; the best-recovery cell in each row is in bold. Cell *n*s are 43, 33, 33, and 38 for Waves 1 and 2, 39, 31, 31, and 34 for Wave 3, and 40, 31, 31, and 37 for the NEI-VFQ measures. Permutation test (5,000 label shuffles): Fast × High GHE best on 11 of 13 outcomes, expected 3.1 under the null, $p = 0.015$.

Comparisons With Belief-Matched Controls

The second analysis asks which treatment participants actually outperformed the active controls. We split the fast healing group into four cells by sensitivity (median split at 2.0) and daily attention (split at 0.35, near the treatment-group median), and compared each cell's mean change in visual acuity at each wave with the mean change among control participants at the same level of sensitivity (high-sensitivity controls, $n = 54$; low-sensitivity controls, $n = 14$). Because control participants also watched a preoperative video and answered the same quiz questions, this comparison holds constant how well participants absorbed the healing-time information they were given and isolates the combination of faster expected timelines with attention to healing. Figure 11 presents the results as raw differences in logMAR change, with positive values indicating more improvement than matched controls. Only one cell clearly and consistently outperformed its matched controls: participants who were both sensitive to the video and attentive to healing (*n* = 22) improved by an additional 0.042 logMAR at Wave 1, 0.040 at Wave 2, and 0.046 at Wave 3 relative to high-sensitivity controls – roughly half a Snellen line at every wave. The other three cells were at or below their matched controls at every wave (high sensitivity with low attention: -0.010, -0.032, -0.030; low sensitivity with high attention: -0.064, -0.028, -0.070; low sensitivity with low attention: -0.063, +0.004, -0.008).

Figure 11

Visual acuity improvement relative to sensitivity-matched controls



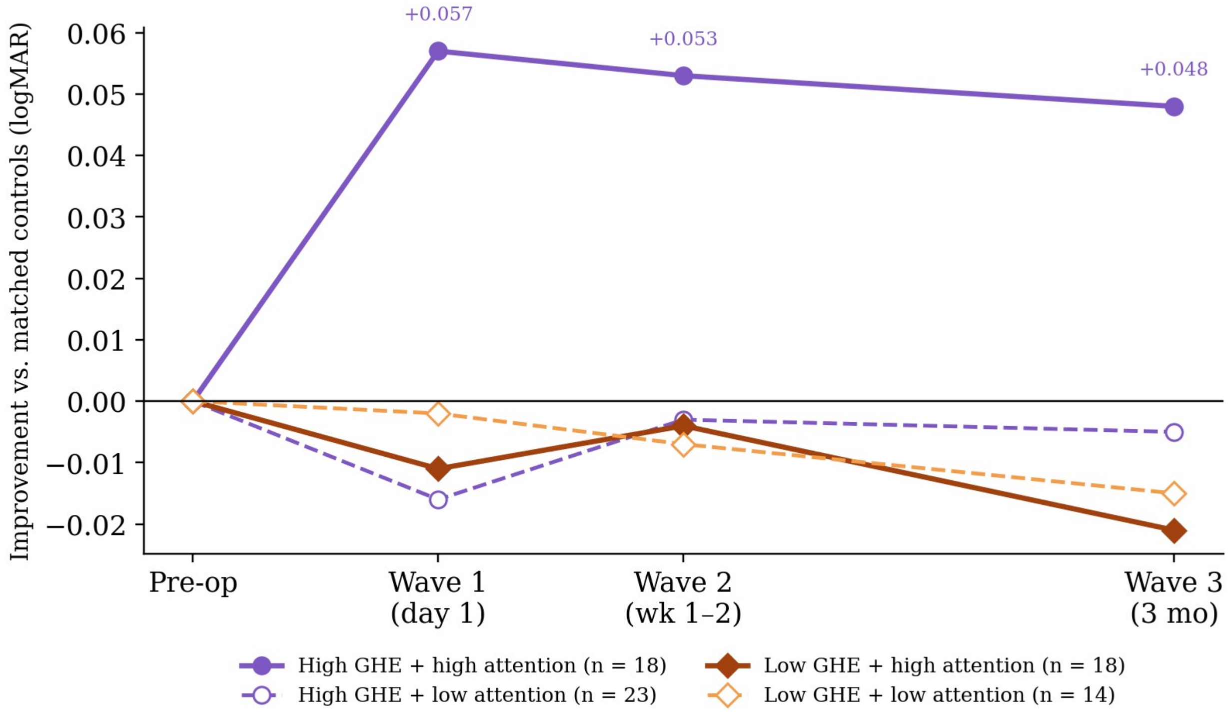
Note. Each line is the mean change in logMAR from baseline for a fast healing subgroup minus the mean change for control participants at the same sensitivity level (high-sensitivity controls, *n*

= 54; low-sensitivity controls, *n* = 14). Positive values indicate greater improvement than matched controls. Median splits: sensitivity at 2.0, daily attention at 0.35. Legend *n*s are subgroup sizes at Waves 1 and 2; Wave 3 *n*s are 20, 19, 12, and 14 (in legend order) owing to attrition and the exclusion of one patient with a retinal detachment. Wave spacing is schematic.

The same pattern emerged when we split treatment participants by General Healing Expectations instead of sensitivity and compared each cell with GHE-matched controls (high-GHE controls, *n* = 33; low-GHE controls, *n* = 38; Figure 12). Participants with high GHE and high attention (*n* = 18) outperformed matched controls by 0.057, 0.053, and 0.048 logMAR at Waves 1, 2, and 3, whereas the remaining three cells were indistinguishable from or slightly below their matched controls (all differences between -0.021 and -0.002). Across both operationalizations of belief, then, the fast healing intervention outperformed an active control only for participants in whom favorable beliefs and attention to healing were both present – the conjunction the EMH framework identifies as the engine of mind-body effects on health.

Figure 12

Visual acuity improvement relative to GHE-matched controls



Note. Each line is the mean change in logMAR from baseline for a fast healing subgroup minus the mean change for control participants at the same level of General Healing Expectations (high-GHE controls, *n* = 33; low-GHE controls, *n* = 38). Positive values indicate greater improvement than matched controls. Median splits: GHE at 4.67, daily attention at 0.35. Wave 3 *n*s are 16, 21, 17, and 14 (in legend order). Wave spacing is schematic.

We are thus confident that the belief × attention interaction – so central to the EMH framework – is robust and generalizable across moderators, outcomes, and analytic approaches. We chose to lead with Sensitivity because that measure is more closely connected to beliefs and expectations in this specific context, rather than beliefs and expectations in general. The fact that both models converge on a similar pattern strongly supports the belief x attention prediction generated by the EMH framework.

General Discussion

In this study, we sought to extend our perceived time and physical healing findings to recovery following cataract surgery – i.e., from the contrived settings of the laboratory to a real life setting with clear and meaningful implications. More specifically, we tested an intervention designed to accelerate healing after cataract surgery that targeted the pillars of the Embodied Models of Health theoretical framework: we targeted beliefs and expectations with the preoperative video we sent to participants before each surgery, and attention to variability with the daily check-in surveys completed by participants in the fast healing condition. The group-level differences were directionally consistent with our hypothesis but modest in magnitude. Crucially, however, the more theoretically motivated analyses revealed that the intervention worked through the specific mechanisms the EMH framework predicts: the coefficient of the Attention × Sensitivity interaction was clinically meaningful and statistically significant (SUR joint test $\chi^2(2) = 4.94$, $p_1 = 0.042$), and when we grouped participants into high sensitivity versus low sensitivity categories, the slope of the regression for high sensitivity participants was significantly different from zero, suggesting that attention to variability worked for participants who internalized the expected healing times provided in the video – as the EMH framework would predict, since EMH argues attention is a *belief-weighted* mediator between mind and body. Attention weights incoming information according to beliefs; if those beliefs are vague, ambiguous, or conflicting, the ability to systematically weight incoming sensory data is compromised, reducing top-down influences on physiological outcomes.

The relatively modest mean differences between the treatment and active control group overlook the contribution of the interaction, and it is the interaction that captures the prediction of the Embodied Models of Health framework. Indeed, as mentioned in the introduction, the intervention we tested was remarkably minimal (subtle differences in expected healing times, short daily surveys), and participants could easily encounter standard recovery information elsewhere. Nonetheless, the intervention worked as theorized: when participants absorbed the faster expectations (high sensitivity) *and* attended to healing variability, they healed significantly more. In other words, participants who engaged with the protocol as intended benefitted as predicted, and the effect among those participants was strong.

The remainder of this discussion proceeds in four parts. First, we unpack the Attention × Sensitivity interaction in terms of the EMH framework proposed in Aungle, Matta, Chen, et al. (2026), arguing that the pattern of results — in which attention to healing variability predicted faster recovery among participants who internalized the faster expected healing times — provides the first empirical evidence for the claim that attention functions as a belief-weighted mediator between psychological processes and physiological outcomes. Second, we connect the present findings to the perceived time and healing results reported in Aungle and Langer (2023), framing both studies as convergent evidence that temporal expectations shape healing — one tested under tightly controlled laboratory conditions, the other in a real clinical setting. Third, we discuss the clinical implications of these findings for preoperative communication and postoperative care. Finally, we highlight questions that arise from these results and note how they set the stage for future research.

The Attention * Sensitivity Interaction

In Aungle, Matta, Chen, et al. (2026), the authors argue that psychological factors mediate all forms of health, and they provide a framework for thinking about the mechanisms underlying these effects. In their Embodied Models of Health framework, beliefs are the fundamental driver of mind-body effects, but, crucially, this is because beliefs dominate attention allocation and expectation formation. Beliefs about health and performance in general feed through to specific expectations in context. Attention functions as a belief-weighted mediator "between" mind and body, amplifying signals that allow expectations and reality to converge, which is especially relevant to mind-body effects that proceed directly from mental states to physiological outcomes (i.e., effects that are not driven by the role mental states play in shaping behavior and affect). The preceding description is abstract in order to generalize across the varied and complex contexts to which it applies, but we can see how it works in practice by looking at the pattern of results reported here.

When we designed this study, we included multiple choice questions after the preoperative information videos simply to reinforce the information provided in the video; we did not expect many participants to get those answers wrong. It is difficult to know whether and how information is processed and internalized, but it was important that the provided healing times were absorbed by participants in order for the EMH framework to work. We created the sensitivity variable as a rough proxy for whether participants internalized the provided healing times, which, if anything, the variable likely overstates: some participants may have simply recalled the provided healing times without really believing or internalizing them. Nonetheless, we needed a way to evaluate whether the healing times in the video were internalized, and the multiple choice questions allowed us to identify the participants that absorbed the expected healing times enough to get the questions right. When we split the sensitivity variable into two groups – high sensitivity (at least 2 of the 3 healing time questions answered correctly) and low sensitivity (fewer than 2 questions answered correctly) – we observed a pattern of results remarkably consistent with what the EMH framework would predict. Among those who internalized the faster healing times (i.e., among high sensitivity participants), greater attention to healing variability significantly predicted larger improvements in visual acuity (slope = −0.040, p_1 = .021).

Among those who did not internalize the faster healing times (i.e., among low sensitivity participants), attention to variability showed no meaningful relationship with healing (slope = +0.027, *p* = 0.28). This pattern argues against a simpler interpretation of attention to variability, namely that noticing variability is inherently beneficial. The EMH framework predicts this: attention to variability is neither inherently helpful nor harmful, it's directional. It amplifies signals consistent with whatever model is active. If someone is attending to their healing process without a clear expectation of improvement, what are they attending to? Possibly ambiguity, possibly things that concern them, possibly normal post-surgical discomfort that they interpret as signs of slow or problematic recovery.

The specificity of the Attention × Sensitivity interaction also provides evidence against several alternative explanations for the observed effects. First, the pattern cannot be attributed to demand characteristics since the effect of attention on healing emerged only among participants who had internalized the faster expected healing times. If participants were simply responding to perceived experimenter expectations, we would expect attention to predict healing regardless of sensitivity, which it did not. Second, the results are inconsistent with a simple Hawthorne effect, in which mere participation in a study or increased monitoring drives improvement. We employed an active control group design, in which control group participants were engaged with at the same frequency and for the same duration as treatment group participants. Third, the findings are not reducible to traditional placebo responding. The preoperative video alone, without engaged daily attention, did not produce robust group-level differences between the fast healing and standard healing conditions. It was only when both components were present —

when participants believed in faster healing and actively attended to variability consistent with that belief — that the intervention produced meaningfully better outcomes. The necessity of both components, rather than either one in isolation, is what the Embodied Models of Health framework predicts, and it is this specificity that makes the framework's added value clear. Notably, the same conjunction emerged in every way we examined the data: in the Attention × Sensitivity and Attention × GHE interactions on healing and on 3-month outcomes, in the permutation test across all 13 outcomes, and in the comparisons with belief-matched controls, where the only treatment subgroup to outperform the active control at every wave was the one in which favorable beliefs and attention to healing co-occurred.

From Perceived Time to Temporal Expectations

In Aungle and Langer (2023), perceived time drove healing: participants who believed more time had passed healed more, even though the elapsed time was always the same. In the introduction of this paper, we argue that effects of perceived time on physiological outcomes occur because perceived time is a key input to expectation formation. The results reported here support this interpretation: rather than manipulating perceived time, which requires false temporal feedback that is impractical to provide outside the lab, we directly manipulated the temporal expectations that perceived time would shape.

In Aungle and Langer (2023), the perceived time manipulation was direct and the context was artificial (precisely calibrated timers, mild bruises produced by cupping). In this study, the manipulation was looser (information in a short video, daily check-in surveys between follow up visits) and the context was real (patients recovering from bilateral cataract surgery). *The fact that effects emerged through the predicted mechanism even under much noisier conditions suggests the underlying principle is robust.*

Taken together, these two studies constitute convergent evidence from very different methodological designs, one laboratory-based and one field-based, pointing to the same conclusion: temporal expectations about physiological processes shape how those processes actually unfold. In the laboratory study, perceived time was directly manipulated with false temporal feedback, and its effect on wound healing was measured under tightly controlled conditions. In the present study, temporal expectations were manipulated indirectly through preoperative information and reinforced through daily attention to healing variability, and their effects were measured in a real clinical population recovering from a common surgical procedure. That both approaches yielded results consistent with the same underlying principle, despite the markedly different levels of experimental control, strengthens the case that the phenomenon is real and generalizable.

Moreover, the present study extends the earlier laboratory findings in an important way by illuminating the role of attention and its interaction with beliefs. In the original laboratory study (Aungle & Langer, 2023), participants were required to monitor a timer and observe their wounds throughout the experimental session, meaning that attention to the healing process was built into the design. Although attention was not independently manipulated in that study, subsequent analyses reported in Aungle, Matta, Loecher, et al. (2026) revealed that the healing observation survey data could be used to calculate an overall measure of attention to variability (ATV). Critically, that ATV measure interacted with participants' general healing expectations: participants who both believed they heal quickly and attended closely to variability in their wounds healed the most. This interaction between healing beliefs and attention to variability is precisely the joint belief-attention pathway predicted by the EMH framework's computational model (Equation 1 in Aungle, Matta, Loecher, et al., 2026), in which an attention weight modulates the extent to which incoming sensory evidence updates prior beliefs about one's health state.

Because attention was embedded in the laboratory design, however, rather than independently manipulated, the earlier study could not fully disentangle the contributions of temporal beliefs from the contributions of attention. In the present study, attention was explicitly operationalized through the daily check-in surveys, and its interaction with beliefs — captured by the sensitivity variable — was directly tested. The finding that attention to healing variability predicted faster recovery among participants who had internalized faster expected healing times provides the kind of mechanistic specificity that the earlier laboratory study, by design, could not offer. That both studies converge on the same belief-attention interaction, despite employing very different operationalizations and contexts, strengthens the case that the EMH framework captures a genuine and generalizable mechanism.

Clinical implications

These findings carry several implications for clinical practice. Perhaps the most immediate is that preoperative communication matters more than clinicians may realize. The specific temporal framing of recovery information may actively shape recovery trajectories. This connects to the broader literature on surgical expectations, which has demonstrated that preoperative expectation optimization can improve long-term outcomes (Auer et al., 2016; Rief et al., 2017). Our results add something new to this literature by suggesting that temporal expectations specifically, not just general outcome expectations, play a meaningful role in shaping recovery. The distinction matters because temporal expectations are among the easiest aspects of preoperative communication for clinicians to adjust, and the present findings suggest that even subtle shifts in the timeframes provided to patients could have meaningful downstream physiological consequences.

The daily check-in survey paradigm represents a low-cost, scalable postoperative tool, but it comes with an important caveat. Our results indicate that attention-directing surveys without appropriately calibrated expectations are unlikely to help. This means that a “one-size-fits-all” recovery monitoring app or check-in protocol is unlikely to produce consistent benefits. What the present results support instead is an integrated approach, one that first sets temporal expectations through preoperative communication and then directs patients’ attention to variability consistent with those expectations. As articulated in the EMH framework, the expectation needs consistent sensory evidence that attention can amplify in order for the mechanism to operate effectively. Without the expectation, attention has nothing to anchor to; without attention, the expectation lacks the informational reinforcement needed to sustain its influence on physiology.

It is also worth considering the clinical significance of the observed effect sizes. Among high-sensitivity participants, the slope of daily attention to healing on visual acuity improvement was −0.040 logMAR per unit of the attention composite. In ophthalmological terms, a change of 0.1 logMAR corresponds to one full line on a Snellen eye chart, and clinicians generally consider a change of 0.1 logMAR to be clinically significant. An improvement of 0.040 logMAR is thus approximately equivalent to reading 2 of the 5 letters on the next line down of the chart — a meaningful change, particularly given the minimal nature of the intervention. The entire protocol consisted of a short preoperative information video (under 5 minutes long) and daily check-in surveys that took approximately 2 minutes each to complete. That this low-cost intervention could produce measurable improvements in objective visual acuity among participants who engaged with it as intended suggests a favorable cost-benefit ratio, especially in a surgical context where even small accelerations in recovery can improve patient quality of life and reduce healthcare utilization.

Limitations and Directions for Future Research

Several limitations of the present study point to important directions for future work. First, the sensitivity variable (operationalized as the number of healing-time questions answered correctly after viewing the preoperative video) is an imperfect proxy for belief internalization. Answering a multiple-choice question correctly captures whether participants remembered the information, but it does not directly measure whether they truly believed or internalized it. Some participants may have recalled the correct answer without adopting it as a personal expectation, while others who answered incorrectly may nonetheless have shifted their beliefs. Developing methods that more precisely capture whether a belief has been internalized — and the strength of that belief when activated — would allow for more fine-grained testing of the EMH framework's predictions. Approaches from cognitive science and psychophysics that assess belief strength through behavioral proxies, rather than self-report accuracy, may prove especially useful in this regard.

Second, the daily attention composite is based entirely on self-report. While the check-in surveys were designed to direct and measure attention to healing variability, there remains a meaningful gap between the way attention is conceptualized in the EMH framework, as a precision-modulating process that continuously weights incoming sensory information, and our current ability to measure it in applied settings. Future research should explore methods for measuring attention to bodily processes more directly, potentially through ecological momentary assessment, physiological indicators of interoceptive focus (e.g., fNIRS), or experience-sampling approaches that can capture attentional dynamics in real time.

Third, while the active control group design strengthens our ability to rule out Hawthorne effects, the control group's daily surveys asked about feelings and activities – which does not constitute a pure "no attention" control. Rather, it represents attention directed elsewhere. It remains possible that some control group participants spontaneously attended to their healing physiology, which would attenuate the observed group differences. A stronger test of the attention hypothesis would involve a design in which the control group's attention could be more fully directed away from healing processes.

Fourth, we could not fully control what information participants encountered outside the study. Patients may have searched for recovery information online, spoken with friends or family members who had previously undergone cataract surgery, or received additional guidance from other healthcare providers. Any such exposure to standard recovery timeframes could have undermined the temporal expectations we aimed to establish, particularly for participants in the fast healing condition. Future studies might consider assessing participants' external information exposure as a covariate, or employing study designs that minimize the window for competing information sources.

Fifth, the study was powered for the between-group main effect. The Attention × Sensitivity interaction that is our theoretical focus requires substantially larger samples for equivalent power, so the interaction estimates reported here, while consistent across moderators, outcomes, and analytic approaches, warrant replication in a larger, preregistered sample.

Finally, the IRB-constrained subtlety of the temporal manipulation, while ethically appropriate, likely attenuated the group-level effect. The differences in expected healing times between conditions were deliberately small in order to present minimal risk to participants. If future studies were able to test more substantial differences in preoperative temporal information — within ethical bounds — the group-level effects might be considerably larger. Alternatively, researchers might explore manipulations that achieve stronger belief shifts through means other than information alone, such as experiential demonstrations or testimonials from prior patients, which may be more effective at establishing internalized expectations.

Conclusion

The central finding of this study is that the physiology of healing following cataract surgery is sensitive to the temporal expectations patients hold and the attention they direct toward the healing process — and, critically, that these two factors interact in the manner predicted by the Embodied Models of Health framework. Neither faster expected healing times alone nor attention to healing variability alone was sufficient to produce robust effects. It was the combination of internalized temporal expectations and engaged attention to variability that predicted meaningfully faster recovery. This interaction is not merely a statistical detail; it is the core prediction of the EMH framework, which holds that attention functions as a belief-weighted mediator between psychological processes and physiological outcomes. The present results provide the first direct empirical test of this prediction in a clinical population.

This finding extends the principle demonstrated in Aungle and Langer (2023) — that temporal perceptions shape the physiology of healing — from the controlled conditions of the laboratory to a real clinical setting. It does so through the theoretical lens developed in Aungle, Matta, Chen, et al. (2026), which provides a mechanistic account of how beliefs, expectations, and attention jointly influence health. In this sense, the present study serves as both a standalone empirical contribution, demonstrating that a minimal psychological intervention can measurably influence surgical recovery, and as the applied arm of a broader theoretical argument about the mechanisms through which psychological factors shape physiological outcomes.

If even subtle shifts in temporal expectations and modest redirection of attention can measurably influence surgical recovery, this raises important questions about the full scope of psychological influences on health outcomes, as well as the adequacy of our language for describing them. The common framing of a “mind-body connection” implies two separate systems that communicate across some interface. But the results reported here do not describe a signal sent from the mind to the body. They describe a process in which the psychological intervention shaped how the body healed. The expectations participants held and the attention they directed did not transmit information to a separate physiological system; they constituted part of the physiological process itself. This is the distinction between mind-body connection and mind-body unity — a distinction that, as we have argued elsewhere (Aungle & Langer, 2023; Aungle, Matta, Chen, et al., 2026), has important implications for how we study, understand, and intervene on health.

The broader implications of mind-body unity for medical science, clinical practice, and public health remain to be fully articulated. What the present findings make clear is that those implications are grounded not in metaphor but in empirical evidence that even minimal psychological interventions can reach deep enough to shape physical healing.

References

Ader, R., & Cohen, N. (1975). Behaviorally conditioned immunosuppression. *Biopsychosocial Science and Medicine*, *37*(4), 333-340.

Auer, C. J., Glombiewski, J. A., Doering, B. K., Winkler, A., Laferton, J. A. C., Broadbent, E., & Rief, W. (2016). Patients' expectations predict surgery outcomes: A meta-analysis. *International Journal of Behavioral Medicine*, *23*(1), 49–62. https://doi.org/10.1007/s12529-015-9500-4

Aungle, P., & Langer, E. (2023). Physical healing as a function of perceived time. *Scientific Reports*, *13*(1), 22432. https://doi.org/10.1038/s41598-023-50009-3

Aungle, P., Matta, P. M., Chen, D., & Langer, E. (2026). Embodied models of health: a theoretical framework of mind-body unity. https://doi.org/10.31234/osf.io/ubtxw_v2

Aungle, P., Matta, P. M., Loecher, M., & Chen, D. (2026). A Computational Framework of Mind-Body Unity: Formalizing Embodied Models of Health. https://doi.org/10.31234/osf.io/uq934_v1

Beecher, H. K. (1955). The powerful placebo. *Journal of the American Medical Association*, *159*(17), 1602-1606.

Benedetti, F., & Amanzio, M. (1997). The neurobiology of placebo analgesia: from endogenous opioids to cholecystokinin. *Prog Neurobiol*, *52*(2), 109-125. https://doi.org/10.1016/s0301-0082(97)00006-3

Benedetti, F., Maggi, G., Lopiano, L., Lanotte, M., Rainero, I., Vighetti, S., & Pollo, A. (2003). Open versus hidden medical treatments: The patient's knowledge about a therapy affects the therapy outcome. *Prevention & Treatment*, *6*(1), 1a.

Camerone, E. M., Wiech, K., Benedetti, F., Carlino, E., Job, M., Scafoglieri, A., & Testa, M. (2021). 'External timing' of placebo analgesia in an experimental model of sustained pain. *Eur J Pain*, *25*(6), 1303-1315. https://doi.org/10.1002/ejp.1752

Cohen, S., Kamarck, T., & Mermelstein, R. (1983). A global measure of perceived stress. *Journal of health and social behavior*, 385-396.

Crum, A. J., & Langer, E. J. (2007). Mind-set matters: Exercise and the placebo effect. *Psychological Science*, *18*(2), 165-171. https://journals.sagepub.com/doi/10.1111/j.1467-9280.2007.01867.x?url_ver=Z39.88-2003&rfr_id=ori:rid:crossref.org&rfr_dat=cr_pub%20%200pubmed

De Craen, A. J., Kaptchuk, T. J., Tijssen, J. G., & Kleijnen, J. (1999). Placebos and placebo effects in medicine: historical overview. *Journal of the royal Society of Medicine*, *92*(10), 511-515.

de la Fuente-Fernández, R., Ruth, T. J., Sossi, V., Schulzer, M., Calne, D. B., & Stoessl, A. J. (2001). Expectation and dopamine release: mechanism of the placebo effect in Parkinson's disease. *Science*, *293*(5532), 1164-1166. https://doi.org/10.1126/science.1060937

Enck, P., Bingel, U., Schedlowski, M., & Rief, W. (2013). The placebo response in medicine: minimize, maximize or personalize? *Nature reviews Drug discovery*, *12*(3), 191-204.

Finniss, D. G., Kaptchuk, T. J., Miller, F., & Benedetti, F. (2010). Biological, clinical, and ethical advances of placebo effects. *The Lancet*, *375*(9715), 686-695.

Geuter, S., Koban, L., & Wager, T. D. (2017). The cognitive neuroscience of placebo effects: concepts, predictions, and physiology. *Annual review of neuroscience*, *40*(1), 167-188.

Gosling, S. D., Rentfrow, P. J., & Swann Jr, W. B. (2003). A very brief measure of the Big-Five personality domains. *Journal of Research in personality*, *37*(6), 504-528.

Grzybowski, A., & Kanclerz, P. (2019). Recent developments in cataract surgery. *Current concepts in ophthalmology*, 55-97.

Hales, A. H. (2024). One-tailed tests: Let's do this (responsibly). *Psychological Methods*, *29*(6), 1209.

Holladay, J. T. (1997). Proper method for calculating average visual acuity. In (Vol. 13, pp. 388-391): SLACK Incorporated Thorofare, NJ.

Kaptchuk, T. J. (1998). Powerful placebo: the dark side of the randomised controlled trial. *The Lancet*, *351*(9117), 1722-1725.

Kaptchuk, T. J., Friedlander, E., Kelley, J. M., Sanchez, M. N., Kokkotou, E., Singer, J. P., Kowalczykowski, M., Miller, F. G., Kirsch, I., & Lembo, A. J. (2010). Placebos without Deception: A Randomized Controlled Trial in Irritable Bowel Syndrome. *PloS one*, *5*(12), e15591. https://doi.org/10.1371/journal.pone.0015591

Kaptchuk, T. J., & Miller, F. G. (2015). Placebo effects in medicine. *New England Journal of Medicine*, *373*(1), 8-9.

Kirsch, I. (1985). Response expectancy as a determinant of experience and behavior. *American psychologist*, *40*(11), 1189.

Langer, E. (2023). *The Mindful Body*. Ballantine Books.

Langer, E. J., Chanowitz, B., Palmerino, M., Jacobs, S., Rhodes, M., & Thayer, P. (1990). Nonsequential development and aging. *Higher stages of human development*, 114-136.

Langer, E. J., Janis, I. L., & Wolfer, J. A. (1975). Reduction of psychological stress in surgical patients. *Journal of Experimental Social Psychology*, *11*(2), 155-165.

Langer, E. J., & Rodin, J. (1976). The effects of choice and enhanced personal responsibility for the aged: a field experiment in an institutional setting. *Journal of personality and social psychology*, *34*(2), 191.

Lasagna, L., Laties, V. G., & Dohan, J. L. (1958). Further studies on the “pharmacology” of placebo administration. *The Journal of Clinical Investigation*, *37*(4), 533-537.

Leventhal, H., Phillips, L. A., & Burns, E. (2016). The Common-Sense Model of Self-Regulation (CSM): a dynamic framework for understanding illness self-management. *Journal of behavioral medicine*, *39*(6), 935-946.

Levine, J., Gordon, N., & Fields, H. (1978). The mechanism of placebo analgesia. *The Lancet*, *312*(8091), 654-657.

Mangione, C. M., Lee, P. P., Pitts, J., Gutierrez, P., Berry, S., Hays, R. D., & Investigators, N.-V. F. T. (1998). Psychometric properties of the National Eye Institute visual function questionnaire (NEI-VFQ). *Archives of Ophthalmology*, *116*(11), 1496-1504.

Matta, P.-M., Baurès, R., Duclay, J., & Alamia, A. (2025). Perceived time drives physical fatigue. *bioRxiv*, 2025.2007.2013.664573. https://doi.org/10.1101/2025.07.13.664573

Matta, P. M., Glories, D., Alamia, A., Baurès, R., & Duclay, J. (2024). Mind over muscle? Time manipulation improves physical performance by slowing down the neuromuscular fatigue accumulation. *Psychophysiology*, *61*(4), e14487.

Moseley, J. B., O'malley, K., Petersen, N. J., Menke, T. J., Brody, B. A., Kuykendall, D. H., Hollingsworth, J. C., Ashton, C. M., & Wray, N. P. (2002). A controlled trial of arthroscopic surgery for osteoarthritis of the knee. *New England Journal of Medicine*, *347*(2), 81-88.

National Health Service. (2025). Cataract surgery. https://www.nhs.uk/tests-and-treatments/cataract-surgery/

Pagnini, F., Phillips, D., Bosma, C. M., Reece, A., & Langer, E. (2015). Mindfulness, physical impairment and psychological well-being in people with amyotrophic lateral sclerosis. *Psychology & Health*, *30*(5), 503-517.

Park, C., Pagnini, F., Reece, A., Phillips, D., & Langer, E. (2016). Blood sugar level follows perceived time rather than actual time in people with type 2 diabetes. *Proceedings of the National Academy of Sciences*, *113*(29), 8168-8170. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4961154/pdf/pnas.201603444.pdf

Pirson, M., Langer, E. J., Bodner, T., & Zilcha-Mano, S. (2012). The development and validation of the Langer mindfulness scale-enabling a socio-cognitive perspective of mindfulness in organizational contexts. *Fordham University Schools of Business Research Paper*.

Price, D. D., Finniss, D. G., & Benedetti, F. (2008). A Comprehensive Review of the Placebo Effect : Recent Advances and Current Thought. *Annual review of psychology*, *59*(1), 565-590. https://doi.org/10.1146/annurev.psych.59.113006.095941

Rahman, S. A., Rood, D., Trent, N., Solet, J., Langer, E. J., & Lockley, S. W. (2020). Manipulating sleep duration perception changes cognitive performance–an exploratory analysis. *Journal of psychosomatic research*, *132*, 109992. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7568839/pdf/nihms-1577746.pdf

Rief, W., Shedden-Mora, M. C., Laferton, J. A., Auer, C., Petrie, K. J., Salzmann, S., Schedlowski, M., & Moosdorf, R. (2017). Preoperative optimization of patient expectations improves long-term outcome in heart surgery patients: results of the randomized controlled PSY-HEART trial. *BMC medicine*, *15*(1), 1-13.

Rodin, J., & Langer, E. J. (1977). Long-term effects of a control-relevant intervention with the institutionalized aged. *Journal of personality and social psychology*, *35*(12), 897.

Wager, T. D., & Atlas, L. Y. (2015). The neuroscience of placebo effects: connecting context, learning and health. *Nature Reviews Neuroscience*, *16*(7), 403-418. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6013051/pdf/nihms969556.pdf

Wolf, S. (1950). Effects of suggestion and conditioning on the action of chemical agents in human subjects—the pharmacology of placebos. *The Journal of Clinical Investigation*, *29*(1), 100-109.

Zigmond, A. S., & Snaith, R. P. (1983). The hospital anxiety and depression scale. *Acta psychiatrica scandinavica*, *67*(6), 361-370. https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1600-0447.1983.tb09716.x?sid=nlm%3Apubmed

Zilcha-Mano, S., & Langer, E. (2016). Mindful Attention to Variability Intervention and Successful Pregnancy Outcomes. *J Clin Psychol*, *72*(9), 897-907. https://doi.org/10.1002/jclp.22294